

**Generation and Biochemical
Characterisation of LIMD1 Constructs for
Investigating Interaction with the E3
Ubiquitin Ligase ARIH1**

Eylul Turkay

MSc in Cancer and Molecular and Cellular Biology

Barts Cancer Institute

Barts and The London School of Medicine and Dentistry

Queen Mary University of London

Supervised by

Professor Dr. Tyson V. Sharp

Dr. Kunal Shah

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Abbreviations

6×His – Hexahistidine tag

AGO2 – Argonaute 2

ARIH1 – Ariadne RBR E3 ubiquitin protein ligase 1

DNA – Deoxyribonucleic acid

E2F – E2 transcription factor

GSK3 α – Glycogen synthase kinase 3 alpha

GST – Glutathione S-transferase

HEK293FT – Human embryonic kidney 293FT cells

HIF-1 α – Hypoxia-inducible factor 1-alpha

HRP – Horseradish peroxidase

HRV 3C – Human rhinovirus 3C protease

IDT – Integrated DNA Technologies

IgG – Immunoglobulin G

IPTG – Isopropyl β -D-1-thiogalactopyranoside

KLD – Kinase, ligase and DpnI

LB – Lysogeny broth

LIM1 – LIM domain 1

LIM2 – LIM domain 2

LIM3 – LIM domain 3

LIMD1 – LIM domain-containing protein 1

LUAD – Lung adenocarcinoma

LUSC – Lung squamous cell carcinoma

MBP – Maltose-binding protein

NEB – New England Biolabs

Ni-NTA – Nickel–nitrilotriacetic acid

NP-40 – Nonidet P-40

NSCLC – Non-small-cell lung cancer

PBS – Phosphate-buffered saline

PBST – Phosphate-buffered saline containing Tween-20

PCR – Polymerase chain reaction

PD-L1 – Programmed death-ligand 1

PGAM5 – PGAM family member 5

pRB – Retinoblastoma protein

PVDF – Polyvinylidene fluoride

RBR – RING-between-RING

SDS–PAGE – Sodium dodecyl sulphate–polyacrylamide gel electrophoresis

TAE – Tris-acetate-EDTA

TPX2 – Targeting protein for Xklp2

VHL – von Hippel–Lindau

Abstract

LIM domain-containing protein 1 (LIMD1) is a tumour suppressor implicated in ARIH1-dependent ubiquitination and degradation of programmed death-ligand 1 (PD-L1), but the LIMD1 region required for ARIH1 association remains unknown. This study tested the hypothesis that ARIH1 interacts with a defined LIMD1 region and aimed to establish a truncation-based biochemical mapping system.

N-terminal, C-terminal and internal LIMD1 fragments were expressed in *Escherichia coli* as 6×His–maltose-binding protein fusion proteins. Constructs were sequence-verified, assessed by immunoblotting and purified using nickel-affinity approaches. Full-length LIMD1 was regenerated by site-directed mutagenesis, lysis and purification conditions were compared, and a sequence-verified FLAG–ARIH1 plasmid was generated. Amylose pull-downs using full-length LIMD1 and constructs 2, 3, 7 and 15 were incubated with untransfected HEK293FT lysate to assess recovery of endogenous ARIH1 and candidate-associated proteins. Tagged proteins were detected across the construct panel, but fragmentation and recovery varied substantially. Sonication produced greater full-length LIMD1 recovery than NP-40-assisted lysis under the conditions tested. In pull-down assays, ARIH1 and AGO2 signals were also present in the bead-only control, PGAM5 was not selectively enriched, and TPX2 detection was uninterpretable. Unequal bait recovery prevented reliable comparison between constructs; therefore, no ARIH1-associated LIMD1 region could be assigned. FLAG–ARIH1 transfection was performed, but expression and reciprocal interaction analysis could not be completed within the experimental timeframe. The study established a sequence-verified construct platform but showed that comparable amounts of intact bait and appropriate bead- and MBP-only controls are required for reliable interaction mapping.

1. Introduction

1.1 Lung Cancer and Loss of Chromosome 3p

Lung cancer is one of the most frequently diagnosed cancers and a leading cause of cancer-related mortality worldwide, with an estimated 2.6 million new cases and 1.9 million deaths in 2024.¹ Non-small-cell lung cancer (NSCLC) accounts for approximately 85% of lung cancers and predominantly comprises lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC).²

Lung tumorigenesis involves genetic and epigenetic changes that activate oncogenic signalling or disable tumour-suppressive pathways.² Frequent and early deletion of chromosome 3p is a characteristic genomic alteration in lung cancer, with the 3p21.3 region harbouring several recurrently lost tumour-suppressor genes.^{3–5}

The chromosome 3 common eliminated region 1 lies within 3p21.3 and contains the gene encoding LIM domain-containing protein 1 (LIMD1).^{3–5} In lung cancer, LIMD1 is affected by loss of heterozygosity, while CpG-island promoter hypermethylation is associated with reduced LIMD1 expression.^{3–5} Functional evidence supporting its tumour-suppressive role comes from *Limd1*-deficient mice, which developed greater tumour burden than wild-type controls following chemical carcinogenesis or oncogenic *KRAS* expression. Thus, these models indicate that LIMD1 deficiency increases susceptibility to tumour development.⁵

1.2 LIMD1 Structure and Context-Dependent Functions

LIMD1 is a 676-amino-acid, approximately 72 kDa member of the Ajuba family of LIM proteins. It comprises an extended N-terminal pre-LIM region and three tandem C-terminal LIM domains (Figure 1.1).^{6,7} LIM domains are zinc-binding protein-interaction modules rather than catalytic domains. LIMD1 therefore acts mainly as a scaffold whose effects depend on its binding partners, cellular localisation and signalling context.⁷ The pre-LIM region is predicted to be largely disordered, but this does not predict the folding or solubility of individual recombinant fragments.⁸

LIMD1 participates in several protein complexes with distinct cellular functions. It interacts with retinoblastoma protein (pRB) to repress E2 transcription factor (E2F)-dependent transcription, promotes Argonaute-dependent microRNA silencing,^{9,10} and scaffolds prolyl hydroxylases with the von Hippel–Lindau (VHL) ubiquitin-ligase complex to facilitate hypoxia-inducible factor 1-alpha (HIF-1 α) degradation.¹¹ More recent work further identified LIMD1 as an Argonaute 2 (AGO2) adaptor that promotes miRNA-target engagement, particularly at weakly seed-paired target sites.⁹ These protein interactions demonstrate that LIMD1 function depends on its partners and cellular context, supporting the use of truncation analysis to map specific interaction regions.

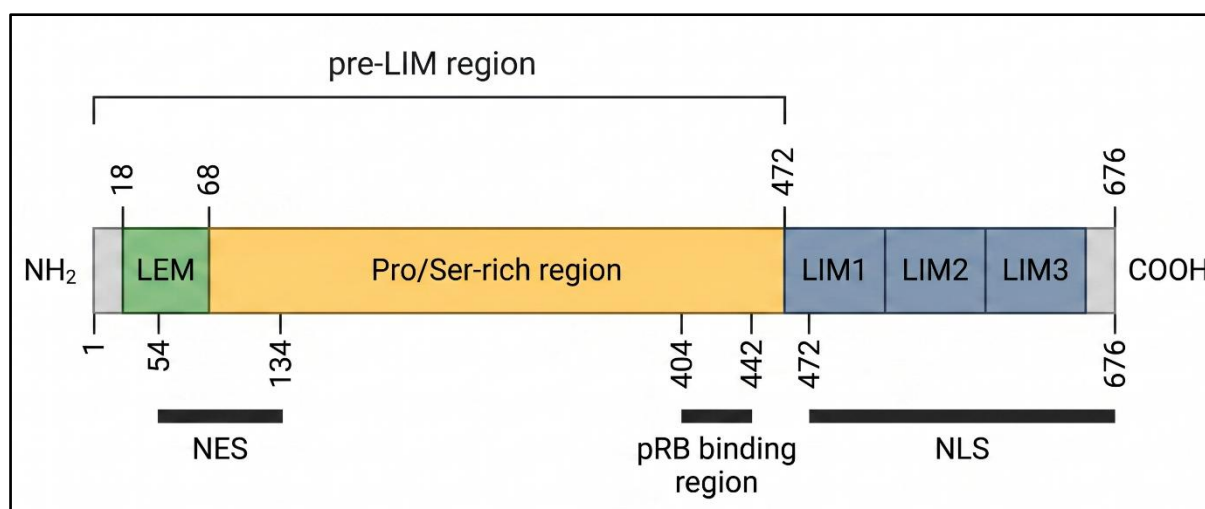


Figure 1.1 Domain architecture of LIMD1. LIMD1 comprises an N-terminal pre-LIM region containing a LEM-like region, nuclear export signal (NES), proline/serine-rich sequences, and a retinoblastoma protein-binding region. Three C-terminal zinc-binding LIM domains provide protein-interaction surfaces and contribute to nuclear localisation. Amino-acid positions are indicated. Adapted from ⁵. Created in BioRender.com.

1.3 LIMD1–ARIH1 Regulation of PD-L1 and Exploratory Candidate Partners

Programmed death-ligand 1 (PD-L1) suppresses antitumour T-cell activity through engagement of programmed death-1. PD-L1 abundance is controlled at several levels including ubiquitination and proteasomal or lysosomal turnover.¹² Ariadne RBR E3 ubiquitin protein ligase 1 (ARIH1) is a RING-between-RING (RBR) E3 ubiquitin ligase that forms a catalytic ubiquitin-thioester intermediate before transferring ubiquitin to its substrate.^{13,14} ARIH1 is normally autoinhibited and can be activated in the context of neddylated cullin-RING ligase complexes.¹⁴ Wu *et al.* (2021) showed that epidermal growth factor receptor inhibition promotes glycogen synthase kinase 3 alpha (GSK3α)-dependent phosphorylation of PD-L1 at Ser279 and Ser283, facilitating ARIH1-mediated PD-L1 ubiquitination and proteasomal degradation.¹³ Catalytically inactive ARIH1 and purified-protein experiments supported a causal role for ARIH1, although the study relied substantially on pharmacological treatment and protein overexpression. Xie *et al.* (2025) independently confirmed that purified ARIH1 can

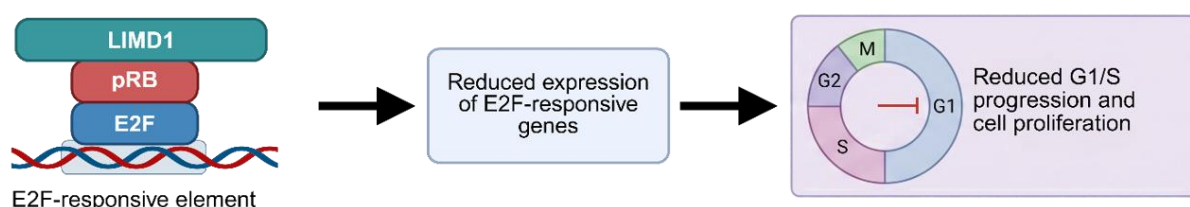
ubiquitinate the PD-L1 cytoplasmic domain.¹⁵ However, comparison with CRL3–SPOP and NEDD4-family ligases indicates that PD-L1 can be regulated by multiple ubiquitin ligases, and ARIH1 should therefore not be assumed to predominate across all cellular contexts.¹⁵

Shah et al. (2026) directly linked LIMD1 to this pathway. In the 421-patient TRACERx cohort, LIMD1 loss was clonal in more than 80% of LUSC and approximately 40% of LUAD. LIMD1 deficiency increased PD-L1 stability and reduced ARIH1-dependent PD-L1 ubiquitination in lung-cancer and airway epithelial models. Endogenous co-immunoprecipitation showed cellular association, while a glutathione S-transferase (GST) pull-down using purified GST-LIMD1 and 6×His–ARIH1 demonstrated direct binding. However, the LIMD1 region responsible for this interaction was not identified.¹⁶ Identifying this region would help define the molecular basis of the LIMD1–ARIH1 interaction.

ARIH1 was the primary candidate in this project. AGO2 was included as a literature-supported comparator because LIMD1 directly participates in Argonaute-dependent microRNA silencing.^{9,10} PGAM5, a mitochondrial stress-signalling phosphatase, and TPX2, a regulator of mitotic-spindle assembly and Aurora-A kinase, were exploratory candidates without an established LIMD1 interaction.^{17,18} Their recovery could therefore indicate preliminary co-enrichment only, not direct binding.

Established LIMD1-containing protein complexes

A LIMD1 -supported repression of E2F-dependent transcription



B LIMD1 scaffolding of PHD2–VHL-mediated HIF-1 α degradation

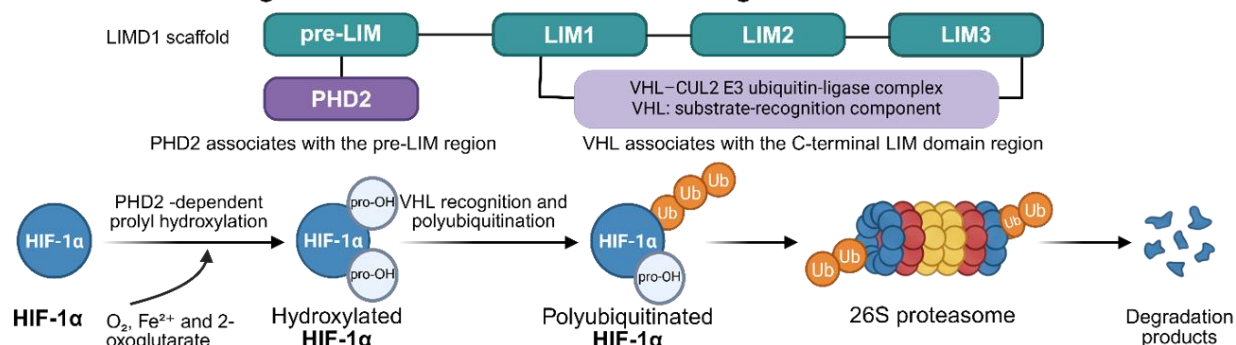


Figure 1.2. Established examples of LIMD1-mediated protein-complex assembly. (A) LIMD1 supports pRB-associated repression of E2F-dependent transcription, restricting cell-cycle progression and proliferation.⁷ (B) LIMD1 scaffolds an association between PHD2 and the VHL–CUL2 E3 ubiquitin-ligase complex, promoting recognition, polyubiquitination and proteasomal degradation of hydroxylated HIF-1 α .⁹ Protein schematics are illustrative and not drawn to scale. Author-created schematic based on ^{11,19}, created using FigureLabs and BioRender.com.

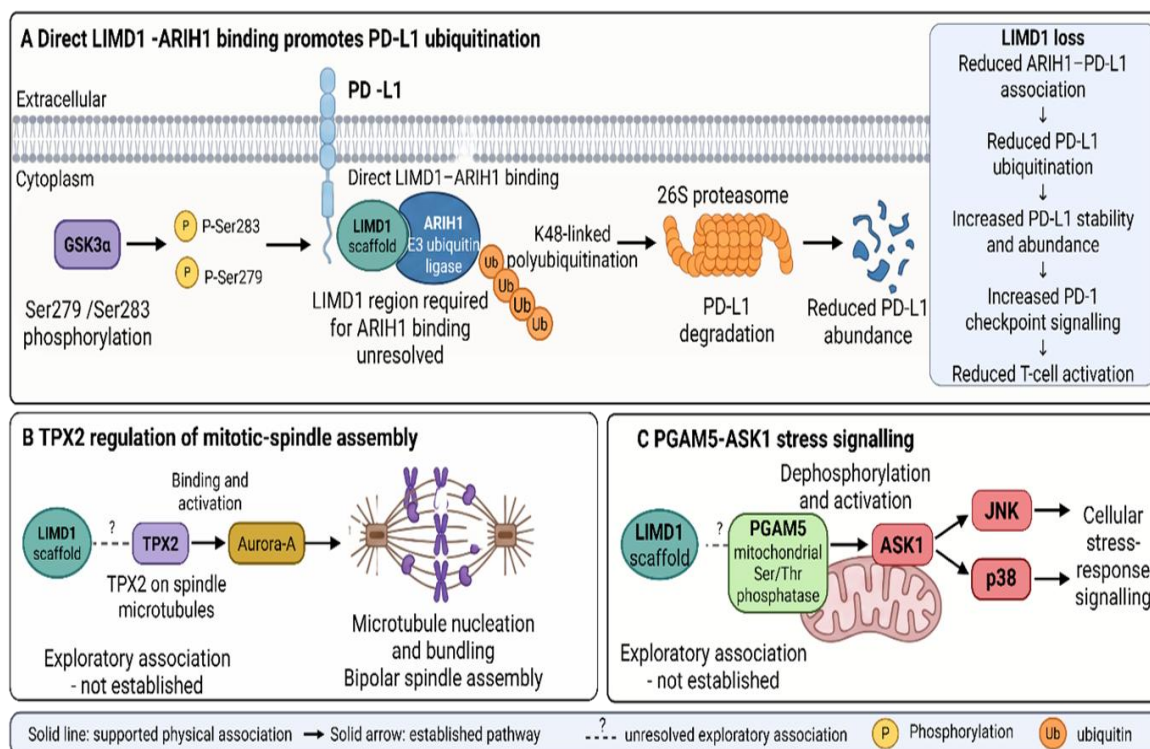


Figure 1.3. Biological context of the LIMD1-ARH1 interaction and exploratory candidate proteins. (A) GSK3α-dependent phosphorylation of PD-L1 at Ser279 and Ser283 promotes ARIH1 recruitment, K48-linked polyubiquitination and proteasomal degradation. LIMD1 directly binds ARIH1 and promotes productive ARIH1-PD-L1 engagement, although the responsible LIMD1 region remains unresolved. (B) TPX2 regulates microtubule organisation, mitotic-spindle assembly and Aurora-A activation. (C) Mitochondrial PGAM5 acts as a protein serine/threonine phosphatase and promotes ASK1-JNK/p38 signalling. TPX2 and PGAM5 were exploratory candidate proteins in this project; their associations with LIMD1 were not established. Solid lines and arrows denote supported associations and pathways, whereas dashed lines and question marks denote unresolved exploratory associations. Protein schematics are illustrative and not drawn to scale. Author-created based on ^{13,16-18}, using FigureLabs and BioRender.com.

1.4 Project Hypothesis and Aim

It was hypothesised that ARIH1 interacts with a defined region of LIMD1, such that deletion of the required region would reduce or abolish ARIH1 recovery in LIMD1-directed pull-down assays.

The primary aim of this study was to identify the LIMD1 region required for ARIH1 association using a recombinant 6×His–MBP–LIMD1 truncation panel. This required validation of the LIMD1 construct panel, optimisation of recombinant expression and affinity purification, and comparison of ARIH1 recovery across selected truncations. A FLAG–ARIH1 construct was additionally generated to enable reciprocal ARIH1-directed affinity capture as a complementary strategy.

2. Materials and Methods

2.1 Key Resources

Table 2.1 Key biological and commercial resources used in this study

Category	Resource	Manufacturer/source	Catalogue number
Cell lines and bacterial strains	HEK293FT cells	Thermo Fisher Scientific	N/A
	<i>E. coli</i> DH5 α competent cells	New England Biolabs (NEB)	C2987I
	<i>E. coli</i> BL21(DE3) competent cells	New England Biolabs (NEB)	C2527I
Plasmids and constructs	pETFP2 expression vector	Sharp Laboratory	N/A
	6 $\times$ His-MBP-LIMD1 truncation constructs	Sharp Laboratory	N/A
	6 $\times$ His-ARIH1 expression plasmid	Sharp Laboratory	N/A
	FLAG-ARIH1 expression plasmid	Generated in this study	N/A
Primary antibodies	Anti-MBP tag mouse monoclonal antibody	Proteintech	66003-1-Ig
	Anti-6 $\times$ His antibody	Abcam	ab18184
	Anti-LIMD1 antibody	Sharp Laboratory	N/A
	Anti-FLAG/DYKDDDDK rabbit monoclonal antibody	Cell Signaling Technology	14793S
	Anti-ARIH1 antibody	Proteintech	14949-1-AP
	Anti-TPX2 antibody	Proteintech	85118-6-RR
	Anti-AGO2 antibody	Sharp Laboratory	N/A
	Anti-PGAM5	Sharp Laboratory	N/A
Secondary antibodies	HRP-conjugated goat anti-mouse IgG	Agilent Dako	P0447
	HRP-conjugated goat anti-rabbit IgG	Agilent Dako	P0448
Affinity purification and pull-down reagents	Amylose Magnetic Beads	New England Biolabs (NEB)	E8035S
	MonoRab TM Anti-DYKDDDDK Magnetic Beads	GenScript	L00835-1
	HisPur TM Ni-NTA Magnetic Beads	Thermo Fisher Scientific	88831
	Ni-NTA Spin Kit	QIAGEN	31314
	Monarch [®] Spin Plasmid Miniprep Kit	New England Biolabs (NEB)	T1110L
DNA purification and molecular biology kits	QIAGEN [®] Plasmid Plus Midi Kit	QIAGEN	12943
	Q5 [®] High-Fidelity DNA Polymerase	New England Biolabs (NEB)	M0493L
	KLD Enzyme Mix	New England Biolabs (NEB)	M0554S

Cell culture/transfection reagents	DharmaFECT™ KB Transfection Reagent	Dharmacon		T-2006-01
	Opti-MEM Reduced Serum Medium	Thermo Scientific	Fisher	31985062
	Dulbecco's Modified Eagle Medium	Thermo Scientific	Fisher	41966029
Protein quantification and detection	SuperSignal™ West Pico PLUS Chemiluminescent Substrate	Thermo Scientific	Fisher	34580
Protein extraction reagents	Lysozyme	Thermo Scientific	Fisher	90082
	Benzonase® Nuclease	Merck Millipore		70664-3
Electrophoresis reagents/standards	GelRed® Nucleic Acid Gel Stain	Biotium		41003
	GeneRuler™ 1 kb Plus DNA Ladder	Thermo Scientific	Fisher	SM1331
	Prestained Protein Ladder - Broad molecular weight (10 - 245 kDa)	Abcam		AB116028

2.2 Bacterial Strains and LIMD1 Constructs

N-terminal, C-terminal and internal LIMD1 truncations were expressed from the pETFPP_2 vector, which encodes an N-terminal 6×His tag, maltose-binding protein (MBP) tag and HRV 3C protease cleavage site upstream of the LIMD1 insert (Figure 2.1). Construct boundaries and predicted molecular masses are summarised in Table 2.2. Constructs 18, 20, 21 and 24 were unavailable.

Plasmids were propagated in *Escherichia coli* DH5α cells, with high-efficiency DH5α used where initial colony recovery was insufficient. Sequence-verified plasmids were transformed into *E. coli* BL21(DE3) for recombinant protein expression.

The experimental workflow is summarised in Figure 2.2.



Figure 2.1 Expression cassette of the pETFPP_2 bacterial expression vector used for recombinant LIMD1 expression. The vector contains an N-terminal 6×His affinity tag (red), maltose-binding protein (MBP) solubility tag (green), and an HRV 3C protease cleavage site (blue) upstream of the cloning site. LIMD1 constructs were inserted immediately downstream of the HRV 3C cleavage site.

Table 2.2. LIMD1 expression constructs used in this study

Construct	LIMD1 residues	Protein region	LIMD1 only (kDa)	+MBP (kDa)	+MBP+6×His (kDa)
1 (regenerated as 15-1)*	1–676	Full-length LIMD1	74.4	116.9	117.0
1	1–676	Full-length LIMD1	74.4	116.9	117.0
2	60–676	N-terminal truncation	67.9	110.4	111.2
3	100–676	N-terminal truncation	63.5	106.0	106.8
4	160–676	N-terminal truncation	56.9	99.4	~100.2
5	250–676	N-terminal truncation	47.0	89.5	90.3
6	350–676	N-terminal truncation	36.0	78.5	79.3
7	465–676	C-terminal LIM region	23.3	65.8	66.6
8	528–676	LIM2–LIM3 region	16.4	58.9	59.7
9	470–533	LIM1 domain	7.0	49.5	50.3
10	532–594	LIM2 domain	6.9	49.4	50.2
11	591–669	LIM3 domain	8.7	51.2	52.0
12	470–594	LIM1–LIM2 domains	13.8	56.3	57.1
13	532–669	LIM2–LIM3 domains	15.2	57.7	58.5
14	470–669	LIM1–LIM3 domains	22.0	64.5	65.3
15	1–465	N-terminal region lacking LIM domains	51.2	93.7	94.5
16	1–80	Extreme N-terminal fragment	8.8	51.3	52.1
17	65–260	Pre-LIM fragment	~21.6	64.1	64.9
19	160–300	Central pre-LIM fragment	15.5	58.0	58.8
22	110–166	Pre-LIM fragment	6.3	48.8	49.6
23	166–260	Pre-LIM fragment	10.5	53.0	53.8

*Predicted molecular masses are shown for the LIMD1 fragment and tagged recombinant proteins. The original full-length Construct 1 was not successfully propagated; full-length LIMD1 was regenerated from Construct 15 by site-directed mutagenesis as Construct 15-1 and designated Construct 1 for subsequent protein experiments.

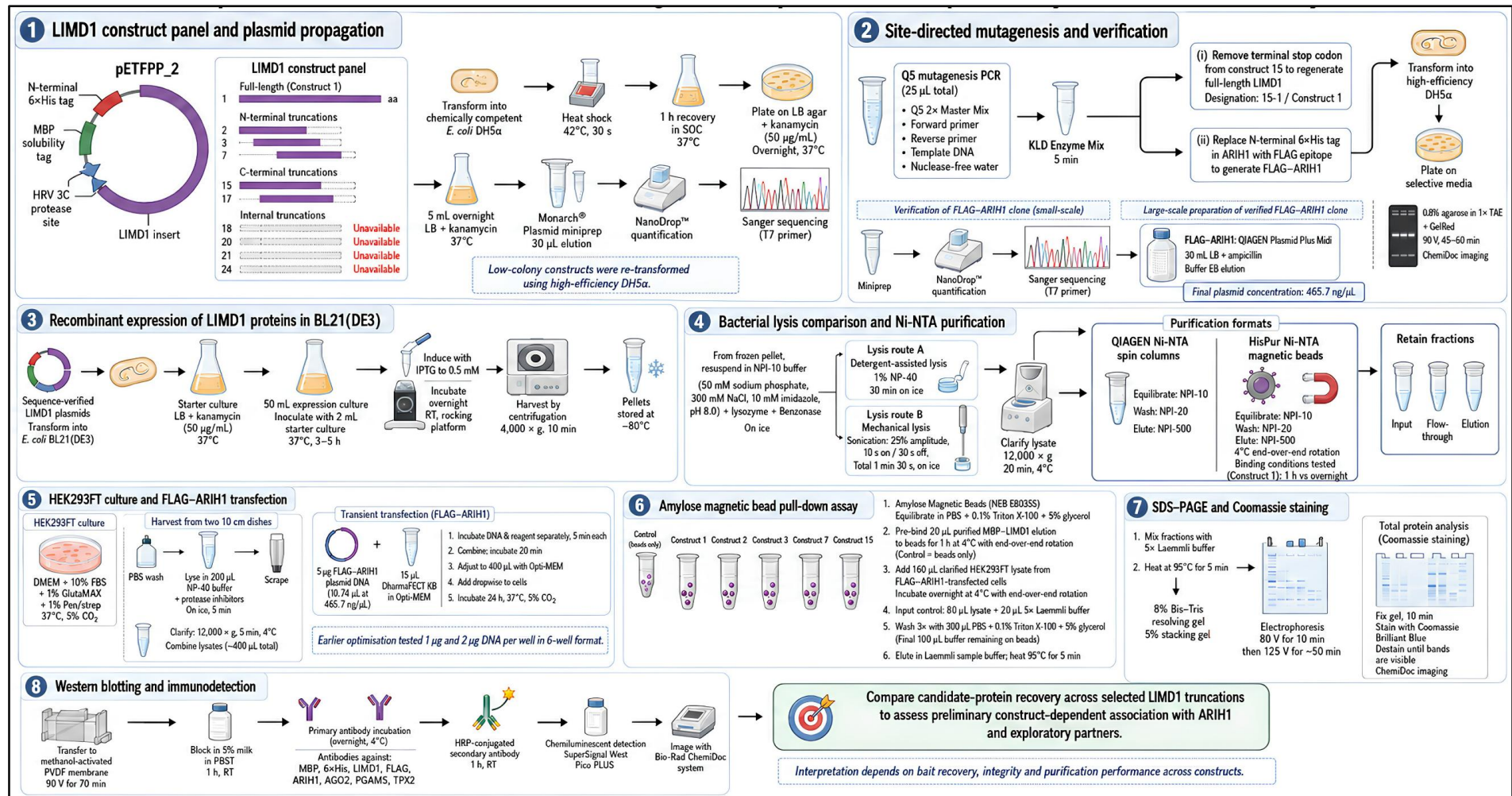


Figure 2.2 Experimental workflow for LIMD1 construct generation, purification and preliminary ARIH1 interaction analysis. The workflow summarises the major experimental stages used in this study, including propagation and sequence verification of the LIMD1 construct panel, site-directed mutagenesis to regenerate full-length LIMD1 and generate FLAG-ARIH1, recombinant LIMD1 expression in *Escherichia coli* BL21(DE3), nickel-affinity purification, HEK293FT transfection, LIMD1-directed amylose affinity pull-down, SDS-PAGE, Coomassie Brilliant Blue staining and immunoblotting. Key experimental conditions, incubation times and sample volumes are indicated. Author-created using BioRender.com and FigureLabs.

2.3 Bacterial Transformation, Plasmid Preparation and Sequence Verification

Plasmids were transformed into chemically competent *E. coli* DH5 α by heat shock at 42°C for 30 s, recovered in SOC medium for 1 h at 37°C and plated on selective LB agar overnight at 37°C. Where required, transformations were repeated using high-efficiency DH5 α cells.

Individual colonies were cultured overnight in 5 mL selective LB medium. Plasmid DNA was purified using the Monarch Plasmid Miniprep Kit, eluted in 30 μ L and quantified using a NanoDrop 2000 spectrophotometer. LIMD1 constructs were verified by Sanger sequencing using a T7 primer.

For large-scale FLAG–ARIH1 plasmid preparation, 500 μ L overnight culture was used to inoculate 30 mL LB containing ampicillin. Following overnight growth at 37°C, plasmid DNA was purified using the QIAGEN Plasmid Plus Midi Kit and quantified by NanoDrop.

2.4 Site-Directed Mutagenesis

Site-directed mutagenesis was performed using Q5 High-Fidelity DNA Polymerase to remove the terminal stop codon from LIMD1 construct 15, generating full-length construct 15-1, and to replace the N-terminal 6 $\times$ His sequence of ARIH1 with a FLAG tag.

PCR reactions were performed in 25 μ L using the primers listed in Appendix Table B1. Products were treated with KLD Enzyme Mix for 5 min, transformed into high-efficiency DH5 α cells and verified by Sanger sequencing.

PCR products were also analysed on 0.8% agarose gels containing GelRed in 1 $\times$ TAE buffer at 90 V for 45–60 min and imaged using a Bio-Rad ChemiDoc system.

2.5 Recombinant Expression of 6×His–MBP–LIMD1 Proteins

Sequence-verified LIMD1 plasmids were transformed into *E. coli* BL21(DE3). Starter cultures were grown in LB containing kanamycin (50 µg/mL). For large-scale expression, 50 mL cultures were inoculated with 2 mL starter culture and grown at 37°C for 3–5 h before induction with 0.5 mM isopropyl β-D-1-thiogalactopyranoside (IPTG). Cultures were incubated overnight at room temperature with agitation, harvested by centrifugation and stored at –80°C.

This approach was adapted from established recombinant LIMD1 expression methods, with modifications for the 6×His–MBP expression system.^{16,19}

2.6 Native Ni-NTA Spin-Column Purification

Bacterial pellets were resuspended in NPI-10 buffer containing 50 mM sodium phosphate, 300 mM NaCl and 10 mM imidazole (pH 8.0), supplemented with lysozyme and Benzonase. Lysates were clarified at 12,000 × *g* and applied to equilibrated QIAGEN Ni-NTA spin columns.

Columns were washed twice with NPI-20 buffer and recombinant proteins eluted with NPI-500 buffer. Input, flow-through and elution fractions were retained for SDS–PAGE, Coomassie staining and immunoblotting.

2.7 Comparison of Bacterial Lysis Methods

The regenerated full-length LIMD1 construct (15-1; subsequently designated Construct 1) was used to compare detergent-assisted and mechanical lysis. Bacterial pellets were resuspended in NPI-10 containing lysozyme and Benzonase.

For detergent lysis, NP-40 was added to 1% (v/v) and samples incubated on ice for 30 min. For mechanical lysis, samples were sonicated on ice at 25% amplitude using

10-s ON/30-s OFF cycles for a total pulse-on time of 90 s. Clarified lysates were subsequently subjected to nickel-affinity purification.

2.8 Magnetic Nickel Affinity Bead Purification

Clarified lysates were incubated with equilibrated HisPur Ni-NTA Magnetic Beads at 4°C with end-over-end rotation. Initial full-length LIMD1 purification used a 1-h binding period, whereas subsequent purifications used overnight incubation.

Beads were collected magnetically, washed four times with NPI-20 and bound proteins eluted with NPI-500. Input, flow-through and elution fractions were analysed by SDS–PAGE, Coomassie staining and immunoblotting.

2.9 HEK293FT Cell Culture

HEK293FT cells were maintained in DMEM supplemented with 10% foetal bovine serum, 1% GlutaMAX and 1% penicillin–streptomycin at 37°C and 5% CO₂. Cells were passaged using trypsin–EDTA at approximately 80–90% confluence.

2.10 Transient Expression of FLAG–ARIH1 in HEK293FT Cells

HEK293FT cells were transfected with sequence-verified FLAG–ARIH1 plasmid using DharmaFECT KB. FLAG–ARIH1 DNA (5 µg; 10.74 µL at 465.7 ng/µL) and 15 µL DharmaFECT KB were separately diluted in Opti-MEM for 5 min, combined for 20 min and adjusted to a final volume of 400 µL before addition to cells.

After 24 h, cells were washed with PBS and lysed in 200 µL NP-40 lysis buffer containing protease inhibitors for 5 min on ice. Lysates were collected by scraping and clarified at 12,000 × g for 5 min. FLAG–ARIH1 was transfected for planned reciprocal affinity capture, but protein expression and reciprocal pull-down analysis were not completed.

2.11 LIMD1-Directed Amylose Affinity Pull-Down

Amylose pull-down assays used recombinant 6×His–MBP–LIMD1 constructs 1, 2, 3, 7 and 15. Amylose Magnetic Beads were equilibrated in PBS containing 0.1% Triton X-100 and 5% glycerol. For each reaction, 15 µL settled beads were washed with 300 µL binding buffer, followed by addition of 20 µL purified LIMD1 elution. LIMD1 was immobilised on the beads for 1 h at 4°C with rotation, and a bead-only control was processed in parallel.

Clarified untransfected HEK293FT whole-cell lysate (160 µL) was then added and samples were incubated overnight at 4°C with rotation. Beads were washed three times with 300 µL binding buffer, and bound proteins were recovered in Laemmli sample buffer at 95°C for 5 min. Samples were analysed by SDS–PAGE and immunoblotting for ARIH1, AGO2, PGAM5 and TPX2, with anti-MBP and anti-6×His used to assess LIMD1 bait recovery.

The assay extended the previously demonstrated direct interaction between full-length LIMD1 and ARIH1 by testing candidate-protein recovery across selected LIMD1 truncations.¹⁶

2.12 SDS–PAGE and Coomassie Brilliant Blue Staining

Protein samples were mixed with 5× Laemmli buffer, heated at 95°C for 5 min and resolved on 8% resolving/5% stacking gels. Electrophoresis was performed at 80 V for 10 min followed by 125 V for approximately 50 min.

For total-protein visualisation, gels were fixed for 10 min, stained with Coomassie Brilliant Blue, destained and imaged using a Bio-Rad ChemiDoc system.

2.13 Western Blotting and Immunodetection

Proteins were transferred to methanol-activated PVDF membranes in Tris–glycine buffer containing 10% methanol at 90 V for 70 min. Membranes were blocked in 5% non-fat milk in PBST for 1 h at room temperature and incubated with primary antibodies (Table 2.3) overnight at 4°C.

Following PBST washes, membranes were incubated with the appropriate HRP-conjugated secondary antibody at 1:10,000 for 1 h at room temperature. Chemiluminescence was developed using SuperSignal West Pico PLUS and imaged using a Bio-Rad ChemiDoc system.

Table 2.3. Working dilutions of antibodies used for the principal immunoblot analyses.

Antibody	Dilution used
Anti-MBP	1:2,000
Anti-6×His	1:2,000
Anti-LIMD1	1:1,000
Anti-ARIH1	1:1,000
Anti-AGO2	1:1,000
Anti-PGAM5	1:1,000
Anti-TPX2	1:1,000
Goat anti-mouse IgG-HRP	1:10,000
Goat anti-rabbit IgG-HRP	1:10,000

3. Results

3.1 Generation and Validation of the LIMD1 Construct Panel

Of the 24 LIMD1 constructs designed for this study, constructs 18, 20, 21 and 24 were unavailable and were therefore not investigated. Construct 1 failed to produce colonies following initial bacterial transformation, while constructs 2, 6, 16, 17, 19, 22 and 23 required retransformations using high-efficiency DH5 α cells before successful plasmid propagation. Transformation outcomes and plasmid DNA concentrations are summarised in Table 3.1.

Site-directed mutagenesis of construct 15 generated the full-length LIMD1 construct 15-1. For consistency with the original construct panel, 15-1 was subsequently designated Construct 1 in protein-expression, purification and pull-down experiments and figures.

Table 3.1. Transformation outcomes and plasmid DNA concentrations for the LIMD1 construct panel. Constructs requiring retransformation using high-efficiency (HE) *E. coli* DH5 α cells are indicated. Plasmid DNA concentrations were determined following miniprep.

Construct	Transformation Outcome	Miniprep (ng/ μ L)
1	Failed	–
2	Successful (HE retransformation)	92.3
3	Successful	77.4
4	Successful	39.3 (198 after precipitation)
5	Successful	62.7
6	Successful (HE)	105.2
7	Successful	63.3
8	Successful	74.7
9	Successful	57.5
10	Successful	54.6
11	Successful	75.3
12	Successful	77.8
13	Successful	109.5
14	Successful	115.3
15	Successful	104.0
16	Successful (HE retransformation)	120.8
17	Successful (HE retransformation)	101.9
19	Successful (HE retransformation)	102.1
22	Successful (HE retransformation)	75.6
23	Successful (HE retransformation)	86.8

3.2 Recombinant Expression Analysis of the LIMD1 Construct Panel

Anti-6×His and anti-MBP immunoblotting detected tagged recombinant proteins across the LIMD1 construct panel, although band intensity and migration varied between constructs (Figures 3.1–3.3). Constructs 2 and 3 produced prominent bands near their predicted fusion-protein masses, together with several lower-molecular-weight bands. Construct 5 showed weak signal at its expected mass, with the strongest band migrating substantially lower. Constructs 6–10 produced prominent bands consistent with their predicted molecular weights, with constructs 9 and 10 migrating at similar positions.

Among constructs 11–23, construct 22 showed comparatively weak signal. Constructs 17 and 19 produced prominent bands close to their predicted masses but also showed additional lower-molecular-weight bands. Construct 15 produced multiple bands rather than a single discrete product at its predicted mass. The remaining constructs showed detectable tagged products, although additional bands were present in several lanes.

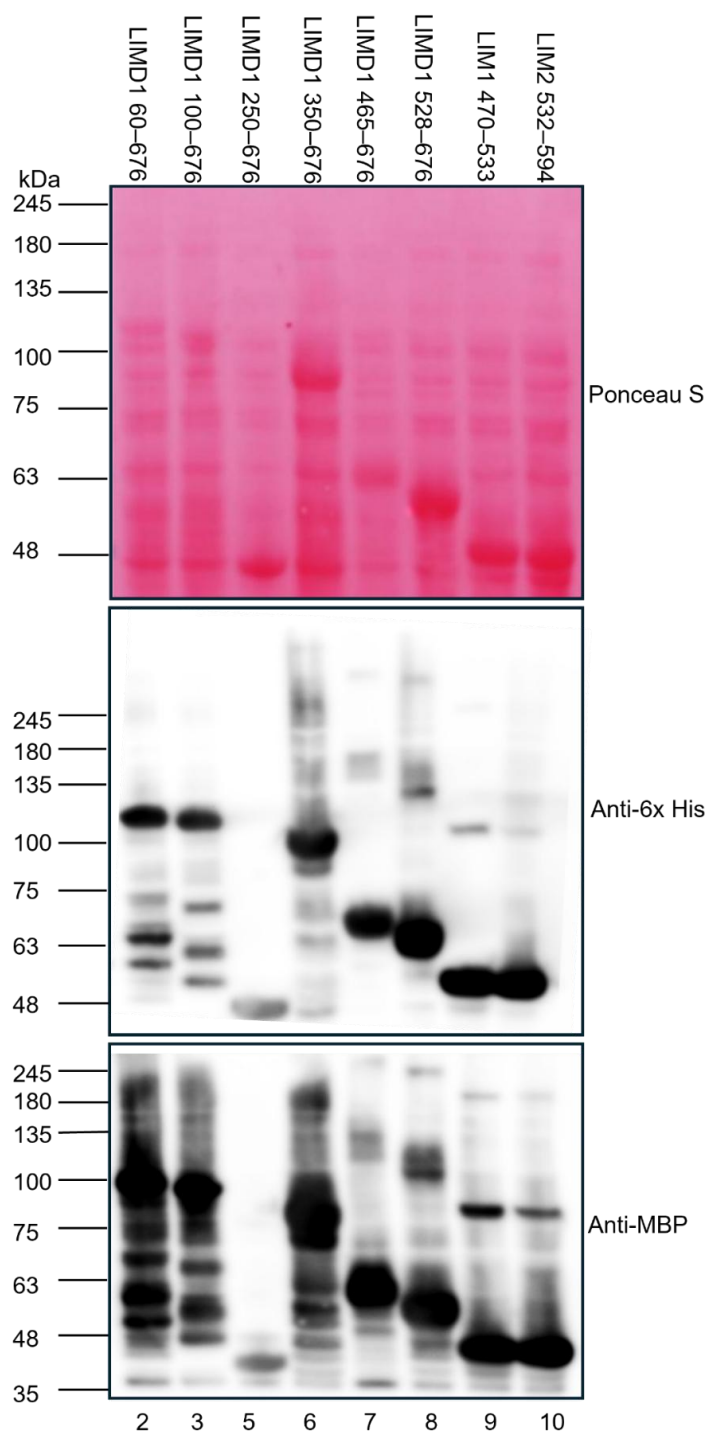


Figure 3.1. Recombinant expression analysis of LIMD1 truncation constructs 2–10. Whole-cell lysates of IPTG-induced 6×His-MBP-LIMD1 constructs expressed in *E. coli* BL21(DE3) were resolved by SDS-PAGE on 8% Bis-Tris gels and transferred to PVDF membranes. Upper panel: Ponceau S staining of transferred protein; middle panel: anti-6×His western blot; lower panel: anti-MBP western blot. Constructs shown are LIMD1 60–676 (2), LIMD1 100–676 (3), LIMD1 250–676 (5), LIMD1 350–676 (6), LIMD1 465–676 (7), LIMD1 528–676 (8), LIM1 470–533 (9) and LIM2 532–594 (10). Construct numbers are indicated below the corresponding lanes and LIMD1 amino-acid boundaries above. Molecular masses are indicated in kDa.

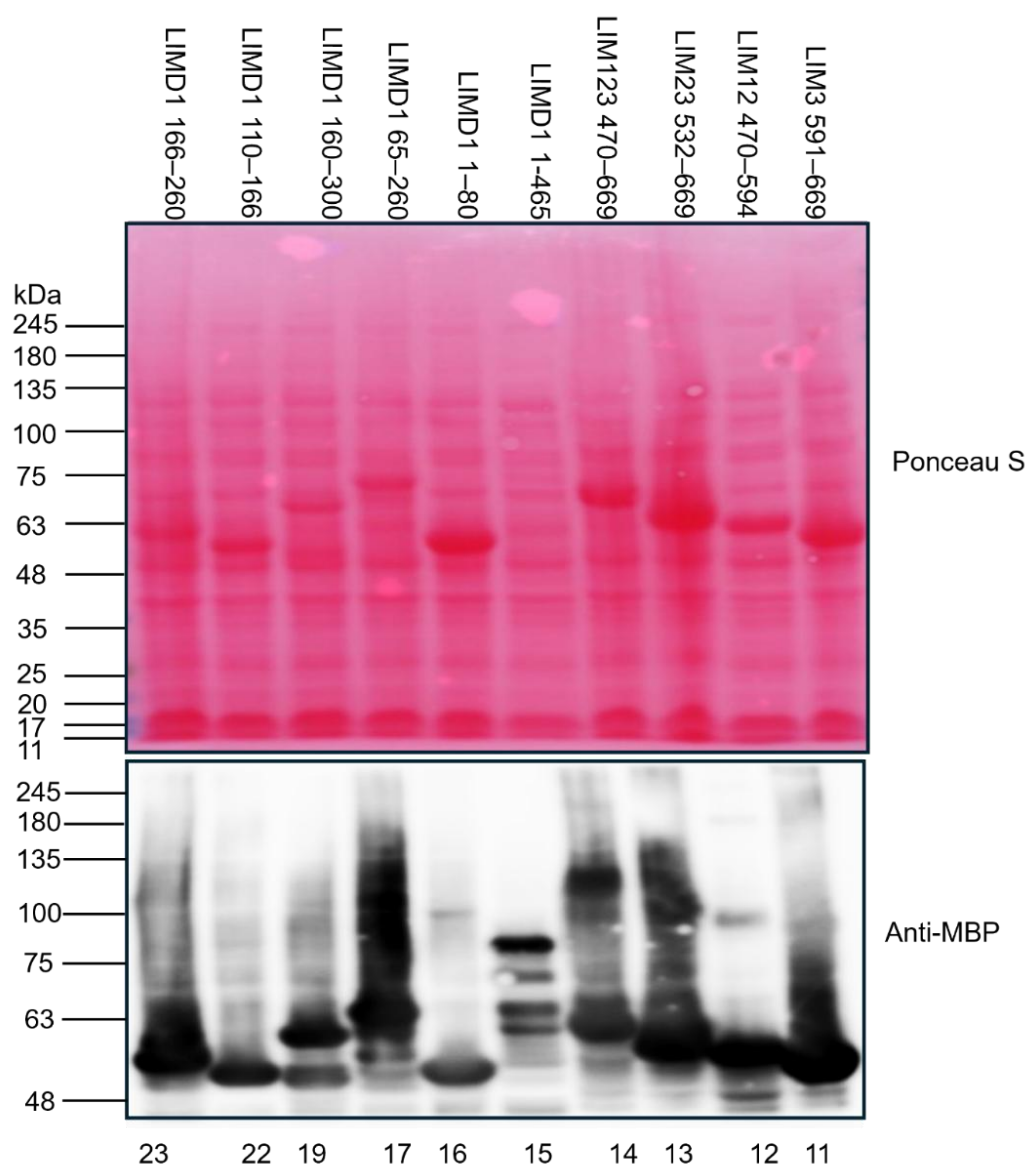


Figure 3.2. Recombinant expression of LIMD1 truncation constructs 11–23 assessed by Ponceau S staining and anti-MBP immunoblotting. Whole-cell lysates of IPTG-induced 6×His-MBP-LIMD1 constructs expressed in *E. coli* BL21(DE3) were resolved by SDS-PAGE on 8% Bis-Tris gels and transferred to PVDF membrane. Upper panel: Ponceau S staining of transferred protein; lower panel: anti-MBP western blot. Constructs shown, from left to right, are LIMD1 166–260 (23), LIMD1 110–166 (22), LIMD1 160–300 (19), LIMD1 65–260 (17), LIMD1 1–80 (16), LIMD1 1–465 (15), LIM123 470–669 (14), LIM23 532–669 (13), LIM12 470–594 (12) and LIM3 591–669 (11). Construct numbers are indicated below the corresponding lanes and LIMD1 amino-acid boundaries above. Molecular masses are indicated in kDa.

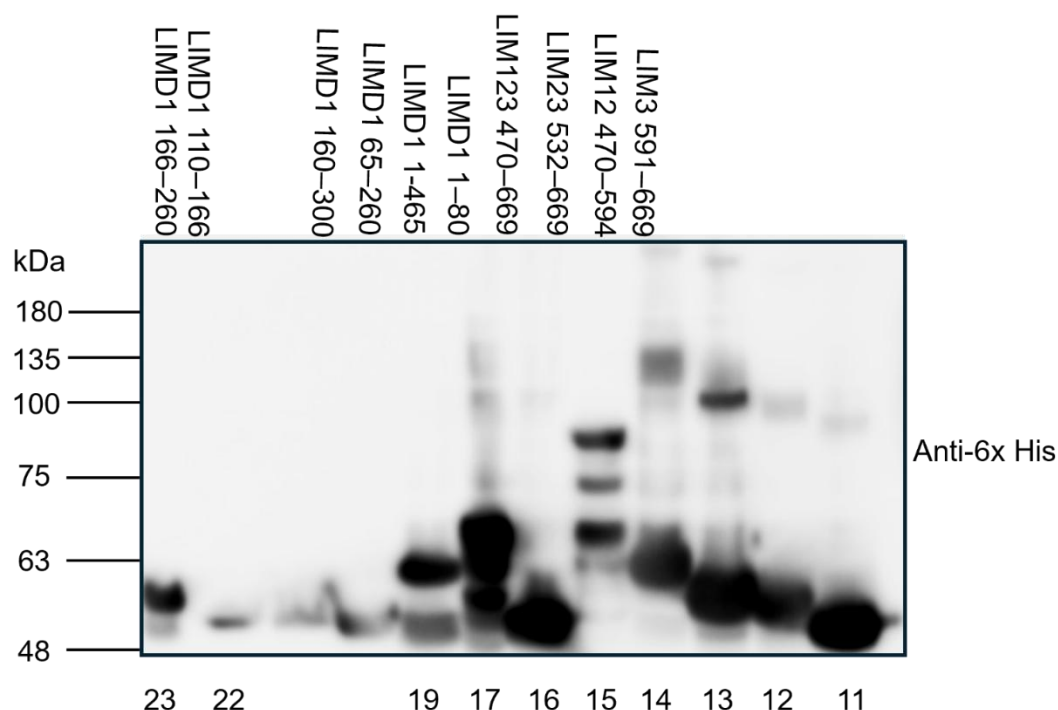


Figure 3.3. Anti-6×His immunoblot analysis of recombinant LIMD1 truncation constructs 11–23. Whole-cell lysates of IPTG-induced 6×His–MBP–LIMD1 constructs expressed in *E. coli* BL21(DE3) were resolved by SDS–PAGE on 8% Bis–Tris gels, transferred to PVDF membrane and analysed by anti-6×His immunoblotting. Constructs shown, from left to right, are LIMD1 166–260 (23), LIMD1 110–166 (22), LIMD1 160–300 (19), LIMD1 65–260 (17), LIMD1 1–80 (16), LIMD1 1–465 (15), LIM123 470–669 (14), LIM23 532–669 (13), LIM12 470–594 (12) and LIM3 591–669 (11). Construct numbers are indicated below the corresponding lanes and LIMD1 amino-acid boundaries above. Molecular masses are indicated in kDa.

3.3 Expression of Full-Length LIMD1 (Construct 15-1) and Expanded Construct Panel

Site-directed mutagenesis of construct 15 generated a whole-plasmid PCR product of approximately the expected size (Figure 3.4). Full-length LIMD1 construct 15-1 was verified by Sanger sequencing (Appendix Figure C1) before recombinant expression. Anti-MBP immunoblotting detected a prominent product in construct 15-1 at approximately 115–120 kDa, consistent with the predicted molecular mass of intact 6×His–MBP–LIMD1 (Figure 3.5). Constructs 2, 3 and 4 also produced prominent products broadly consistent with their predicted fusion-protein masses. In contrast, replicate construct 5 lanes predominantly contained an MBP-reactive species at approximately 48–55 kDa, substantially below the predicted mass of the intact fusion protein. Constructs 6–8 also produced MBP-reactive products with differing signal intensities and migration patterns.

Anti-LIMD1 immunoblotting detected strong reactivity in construct 15-1 and detectable signal in constructs 2–4, whereas little or no detectable signal was observed in constructs 5–8 under the conditions used (Figure 3.5).

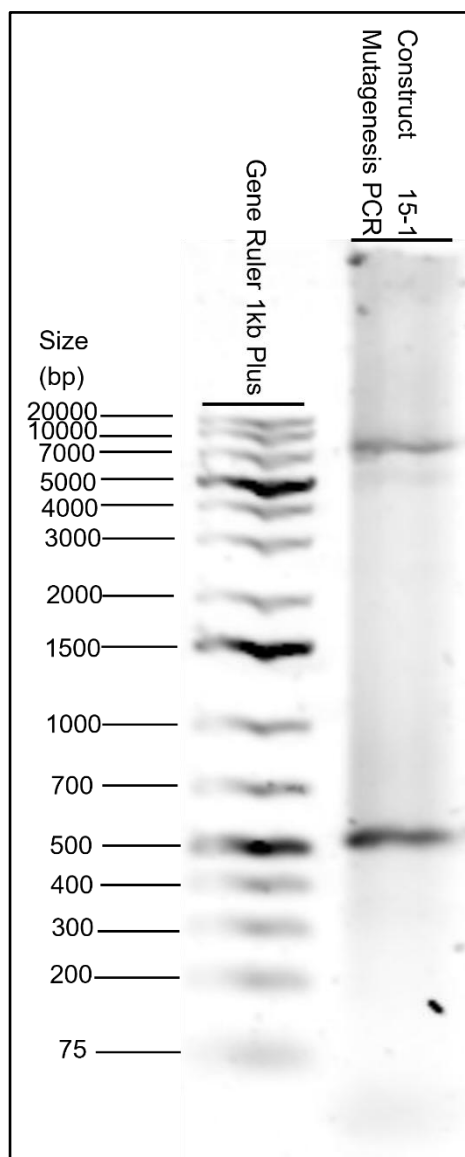


Figure 3.4. Agarose gel electrophoresis of construct 15-1 whole-plasmid mutagenesis PCR. Site-directed mutagenesis PCR was performed on construct 15 to remove the terminal stop codon and generate full-length LIMD1 construct 15-1. The PCR product was resolved by agarose gel electrophoresis alongside a GeneRuler 1 kb Plus DNA Ladder. DNA fragment sizes are indicated in bp.

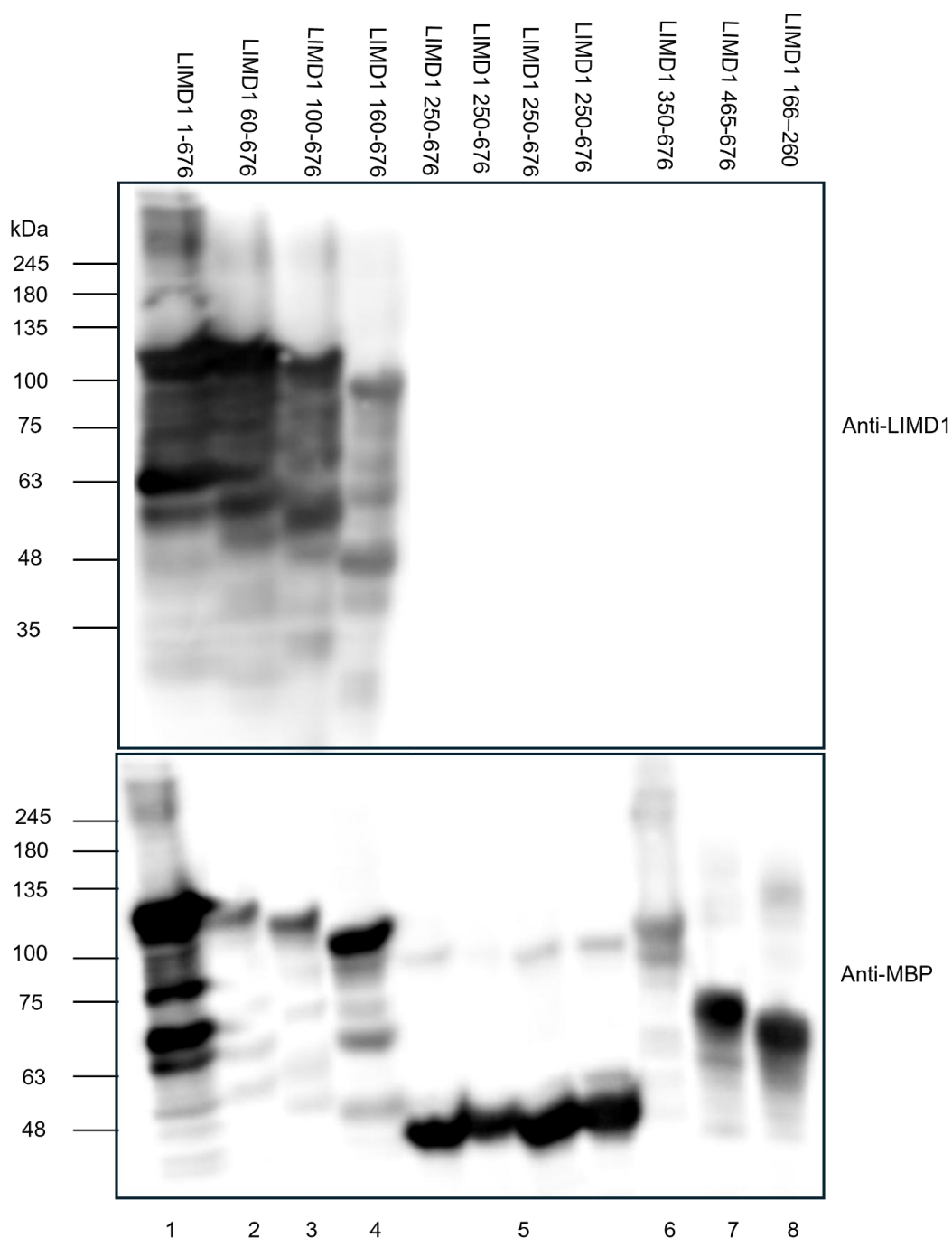


Figure 3.5 Expression of full-length LIMD1 and selected truncation constructs. Whole-cell lysates were resolved by SDS-PAGE on 8% Bis-Tris gels. Upper panel: anti-LIMD1 western blot. Lower panel: anti-MBP western blot. Samples comprise full-length LIMD1 (construct 15-1), LIMD1 60–676 (2), LIMD1 100–676 (3), LIMD1 160–676 (4), LIMD1 250–676 (5; four replicate lanes), LIMD1 350–676 (6), LIMD1 465–676 (7) and LIMD1 528–676 (8). Prestained protein molecular weight marker (10–245 kDa; Abcam) was loaded in the outermost lane.

3.4 Generation of the FLAG–ARIH1 Construct

Initial ARIH1 mutagenesis did not produce the intended replacement of the N-terminal 6×His sequence with the FLAG epitope, based on re-examination of the sequencing chromatogram and absence of anti-FLAG immunoreactivity. Mutagenesis was therefore repeated. Repeat PCR generated a product of approximately 7.1 kb, which was treated with KLD Enzyme Mix and transformed into high-efficiency DH5α cells. Sanger sequencing of colony 1 confirmed replacement of the 6×His sequence with the expected FLAG-tag coding sequence (Figure 3.6; Appendix Figure C2).

Large-scale plasmid preparation of the sequence-verified clone yielded plasmid DNA at 465.7 ng/μL. The sequence-verified FLAG–ARIH1 construct was subsequently used for HEK293FT transfection; however, FLAG–ARIH1 protein expression and reciprocal affinity-capture analysis were not completed within the timeframe.

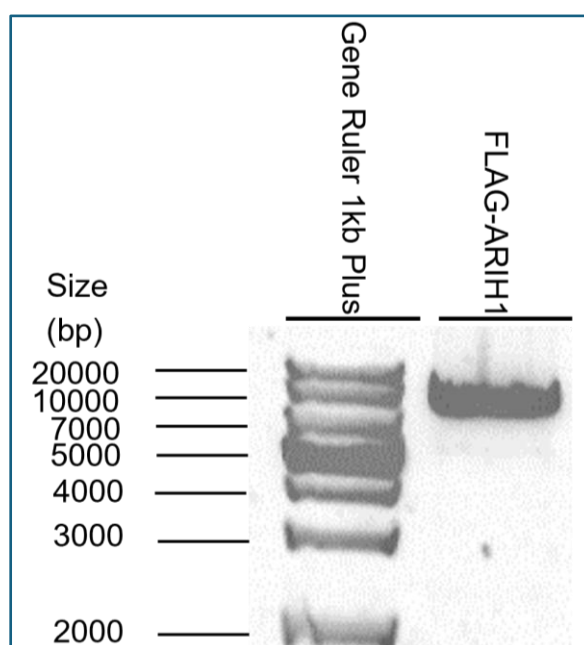


Figure 3.6. Agarose gel electrophoresis of the repeat FLAG-ARIH1 mutagenesis PCR product. Site-directed mutagenesis PCR was repeated following re-evaluation of the original construct. Agarose gel electrophoresis demonstrated amplification of a DNA fragment of approximately 7.1 kb, corresponding to the expected size of the FLAG-ARIH1 expression plasmid. PCR products were resolved on a 0.8% (w/v) agarose gel prepared in 1× TAE buffer using a GeneRuler 1 kb Plus DNA Ladder (Thermo Fisher Scientific) as the molecular size marker.

3.5 Native Ni-NTA Affinity Purification of Recombinant LIMD1 Proteins

Native Ni-NTA spin-column purification was assessed using constructs 2, 3 and 5–10. Input, elution and flow-through fractions were analysed by Coomassie staining and anti-MBP immunoblotting (Figures 3.7 and 3.8).

Recovery varied between constructs. Constructs 2 and 3 showed relatively weak recovery in the elution fractions (Figure 3.7). Construct 5 contained a prominent Coomassie-stained species at approximately 45–50 kDa, substantially below the predicted ~90 kDa mass of the intact fusion protein, with little detectable anti-MBP signal.

Among constructs 7–10, construct 7 showed prominent MBP-reactive protein in the input but comparatively weak recovery in the elution fraction. Constructs 8, 9 and 10 produced detectable MBP-reactive products in their respective elution fractions, with bands migrating close to their predicted fusion-protein masses (Figure 3.8). Full-length LIMD1 was therefore selected for subsequent optimisation of bacterial lysis conditions.

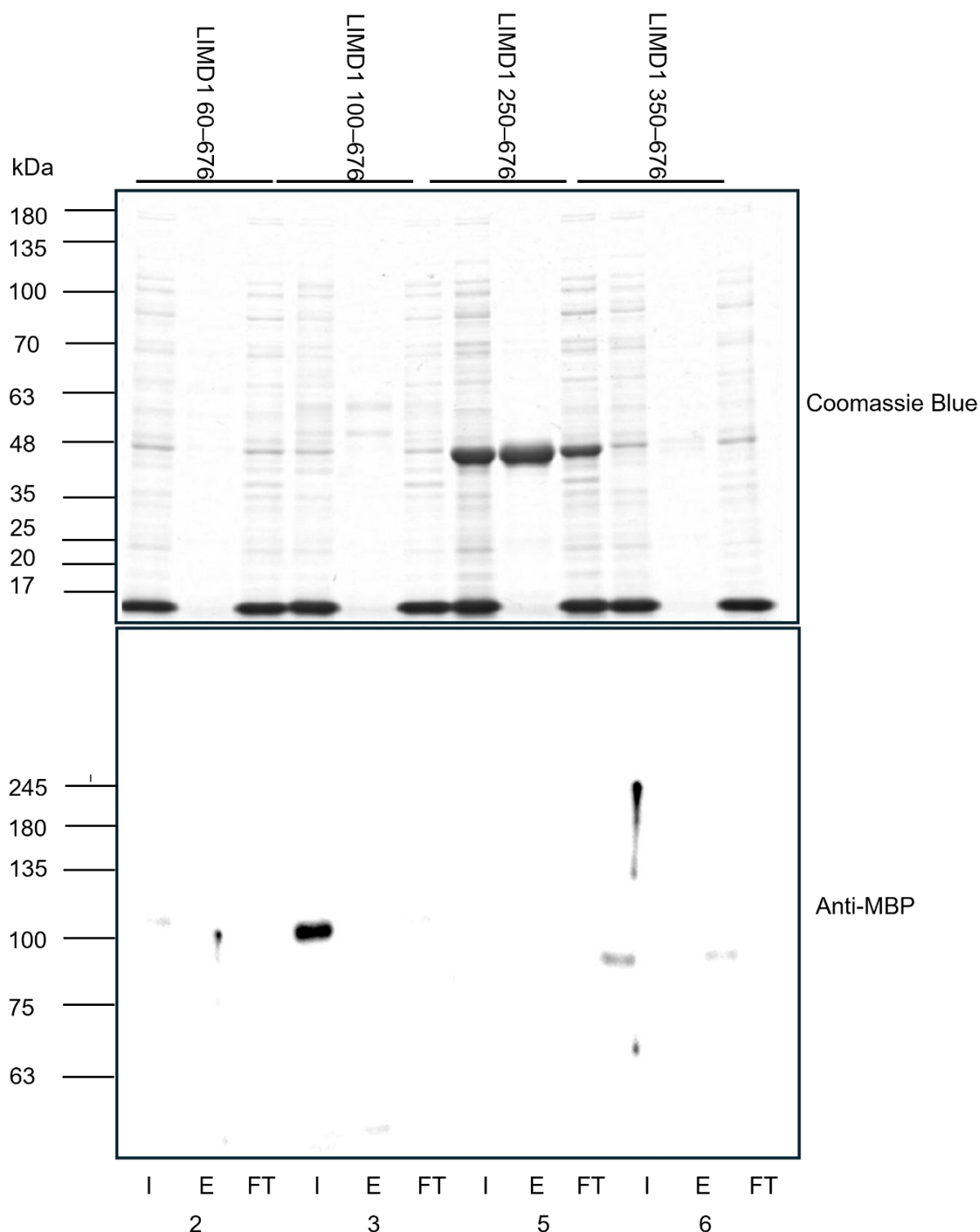


Figure 3.7. Native Ni-NTA affinity purification of recombinant LIMD1 constructs 2–6. Purified protein fractions were resolved by SDS-PAGE on 8% Bis-Tris gels. Upper panel: Coomassie Brilliant Blue-stained gel. Lower panel: anti-MBP western blot of the corresponding fractions. Constructs shown are LIMD1 60–676 (2), LIMD1 100–676 (3), LIMD1 250–676 (5) and LIMD1 350–676 (6). For each construct, Input (I), Elution (E) and Flow-through (FT) fractions are shown following native Ni-NTA spin column purification using NPI-10 lysis buffer and NPI-500 elution buffer. Prestained protein molecular weight marker (10–245 kDa; Abcam) was loaded in the outermost lane.

3.6 Comparison of NP-40 Detergent and Sonication-Assisted Lysis for Full-Length LIMD1 Purification

Following variable recovery during Ni-NTA spin-column purification, full-length LIMD1 was used to compare NP-40- and sonication-assisted bacterial lysis. The sonicated lysate was additionally subjected to Ni²⁺ magnetic-bead purification to assess recovery using an alternative affinity-capture format. Input, elution and flow-through fractions were analysed by Coomassie staining and anti-MBP immunoblotting (Figure 3.9).

Coomassie staining showed abundant protein in the input and flow-through fractions under both lysis conditions, with comparatively lower protein recovery in the elution fractions. Anti-MBP immunoblotting detected full-length MBP-tagged LIMD1 in both input fractions. Following the 1-hour purification incubation, the sonicated preparation showed greater recovery of MBP-reactive protein in the elution fraction than the NP-40-assisted preparation, although substantial signal remained in the flow-through fractions (Figure 3.9).

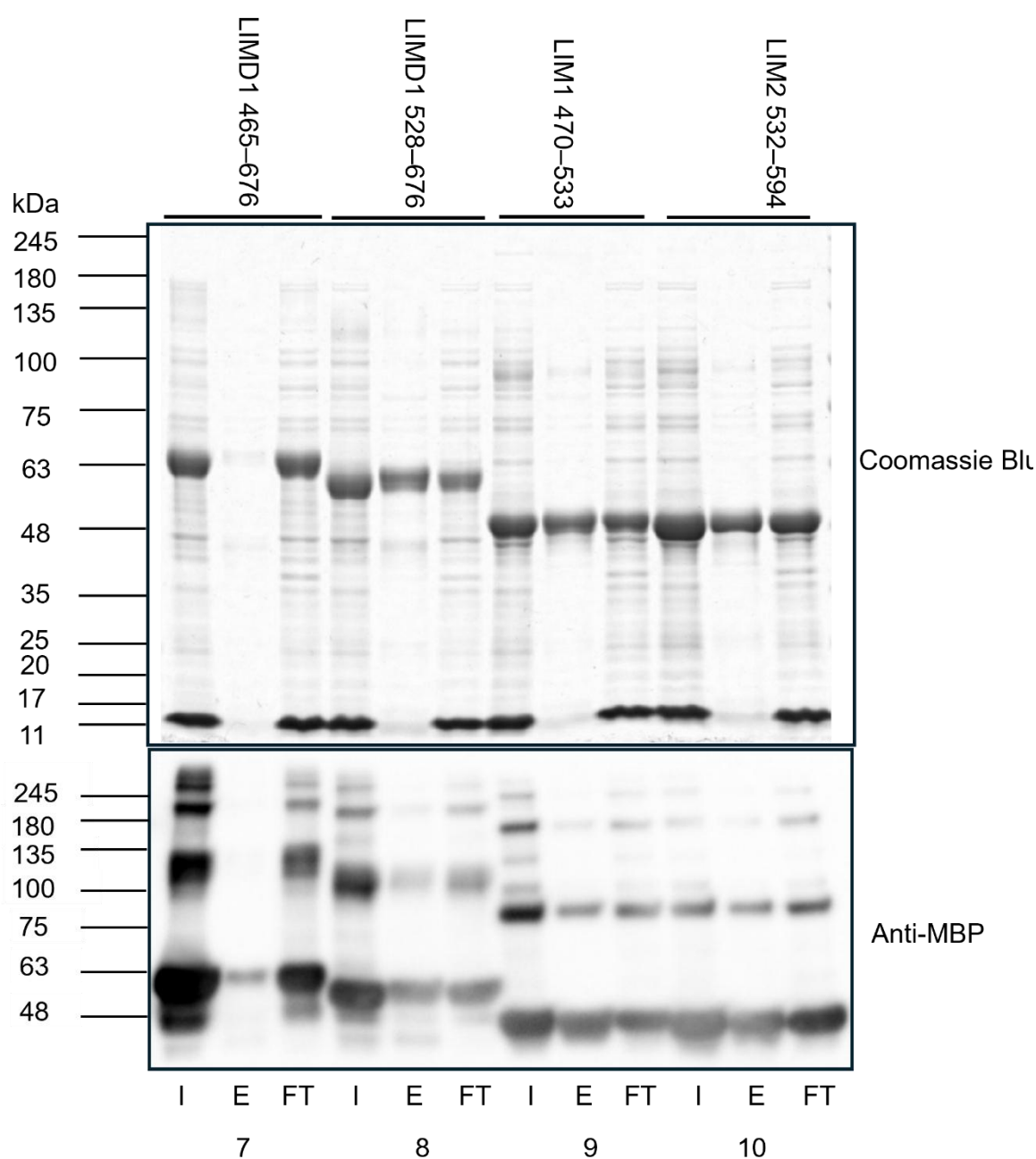


Figure 3.8. Native Ni-NTA affinity purification of recombinant LIMD1 constructs 7–10. Purified protein fractions were resolved by SDS-PAGE on 8% Bis-Tris gels. Upper panel: Coomassie Brilliant Blue-stained gel. Lower panel: anti-MBP western blot of the corresponding fractions. Constructs shown are LIMD1 465–676 (7), LIMD1 528–676 (8), LIM1 470–533 (9) and LIM2 532–594 (10). For each construct, Input (I), Elution (E) and Flow-through (FT) fractions are shown following native Ni-NTA spin column purification using NPI-10 lysis buffer and NPI-500 elution buffer. Prestained protein molecular weight marker (10–245 kDa; Abcam) was used for molecular weight estimation.

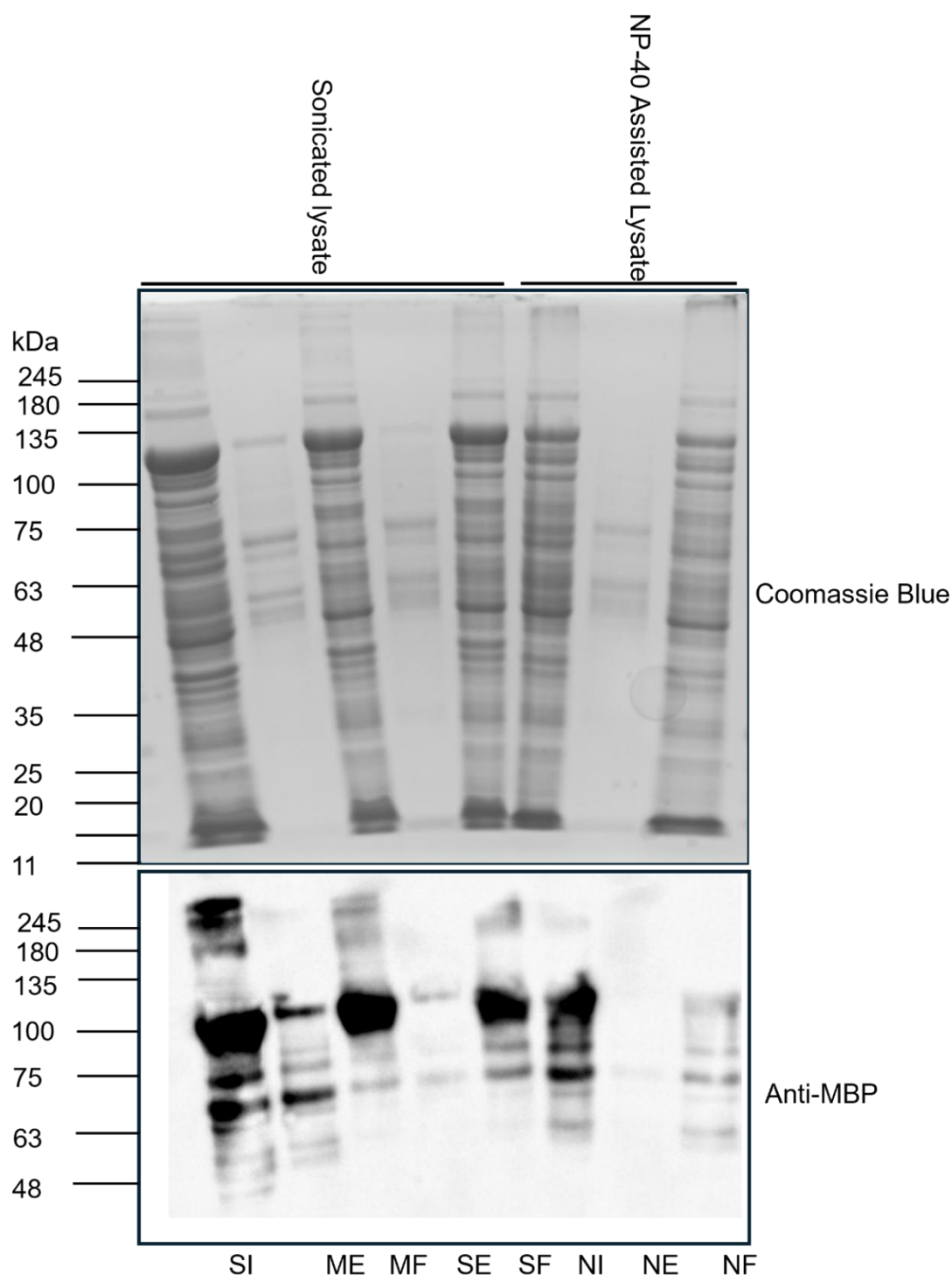


Figure 3.9. Comparison of NP-40 detergent and sonication-assisted lysis for purification of full-length LIMD1. Full-length LIMD1 (Construct 1) was processed using either probe sonication or NP-40-assisted lysis. Protein fractions were resolved by SDS-PAGE on 8% Bis-Tris gels. Upper panel: Coomassie Brilliant Blue-stained gel; lower panel: anti-MBP western blot. SI, sonicated input; ME, Ni^{2+} magnetic-bead elution from the sonicated lysate; MF, Ni^{2+} magnetic-bead flow-through from the sonicated lysate; SE, sonication-associated elution; SF, sonication-associated flow-through; NI, NP-40 input; NE, NP-40 elution; NF, NP-40 flow-through. Molecular masses are indicated in kDa.

3.7 Magnetic Nickel Affinity Bead Purification of Recombinant LIMD1 Proteins

For constructs 1–6, Coomassie-stained SDS–PAGE showed detectable protein in the magnetic-bead elution fractions, although the intensity of recovered protein varied between constructs (Figure 3.10). Construct 5 produced the strongest elution signal by Coomassie staining and anti-MBP immunoblotting, whereas constructs 2 and 3 showed the weakest recovery. Construct 6 produced moderate anti-MBP signal but weak Coomassie staining. Full-length construct 1 produced a clear anti-MBP-reactive elution band comparable to construct 6.

Constructs 15, 16, 22 and 23 also produced detectable elution products (Figure 3.11). Anti-MBP signal was strongest for construct 22, followed by construct 23, intermediate for construct 16 and weakest for construct 15. Coomassie staining showed a similar pattern.

Both constructs 17 and 19 were detected in the input fractions, but only construct 19 produced detectable anti-MBP-reactive protein following elution (Figure 3.12). Construct 17 was therefore excluded from subsequent pull-down experiments. The final purified LIMD1 preparations were compared by Coomassie-stained SDS–PAGE (Figure 3.13).

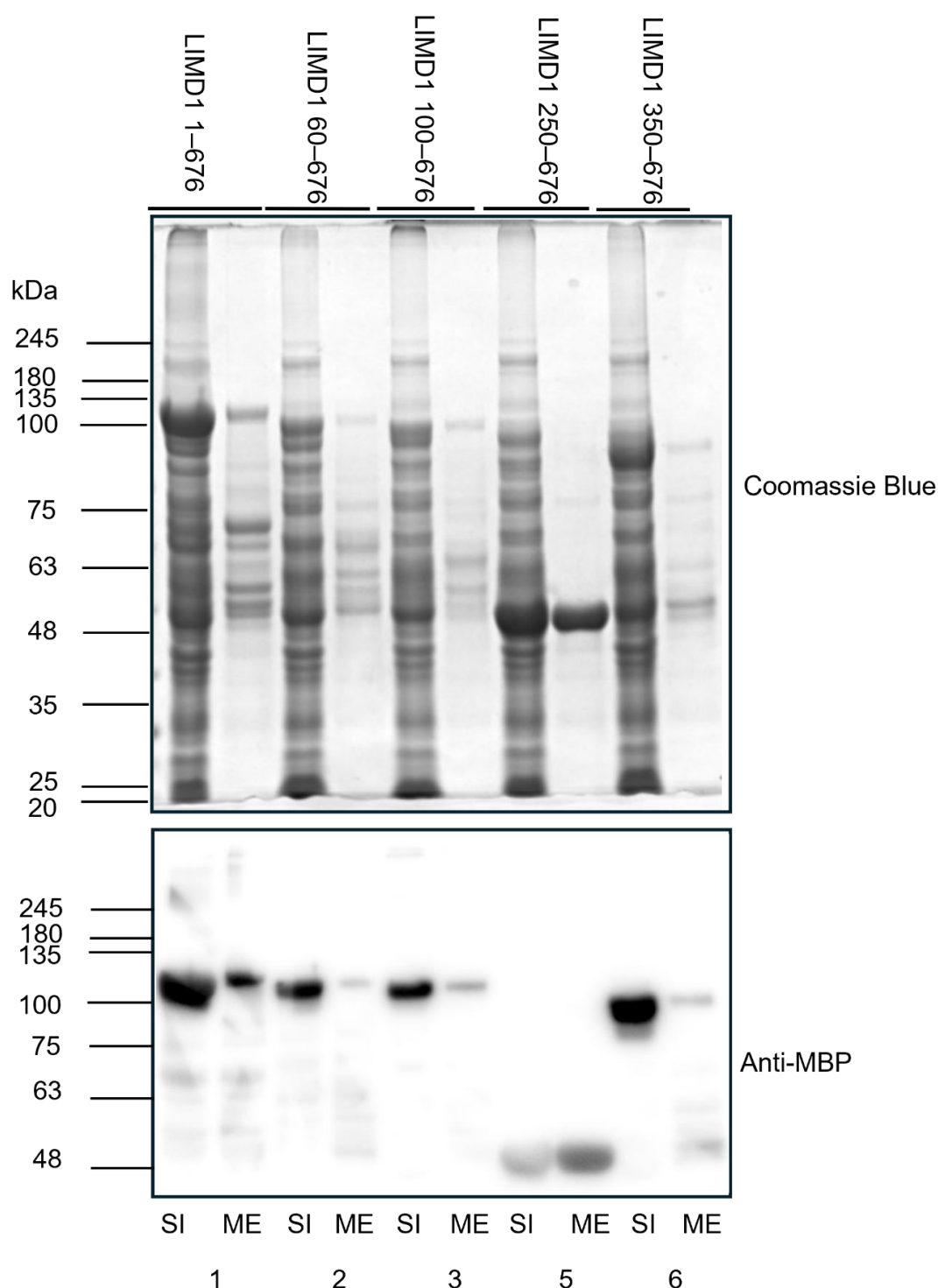


Figure 3.10. Magnetic nickel affinity bead purification of recombinant LIMD1 constructs 1–6. Protein fractions were resolved by SDS-PAGE on 8% Bis-Tris gels. Upper panel: Coomassie Brilliant Blue-stained gel. Lower panel: anti-MBP western blot of the corresponding fractions. Constructs shown are full-length LIMD1 (Construct 1; residues 1–676), LIMD1 60–676 (2), LIMD1 100–676 (3), LIMD1 250–676 (5), and LIMD1 350–676 (6). For each construct, sonicated input (SI) and magnetic bead elution (ME) fractions are shown following purification using magnetic nickel affinity beads under native conditions. Prestained protein molecular weight marker (10–245 kDa; Abcam) was used for molecular weight estimation.

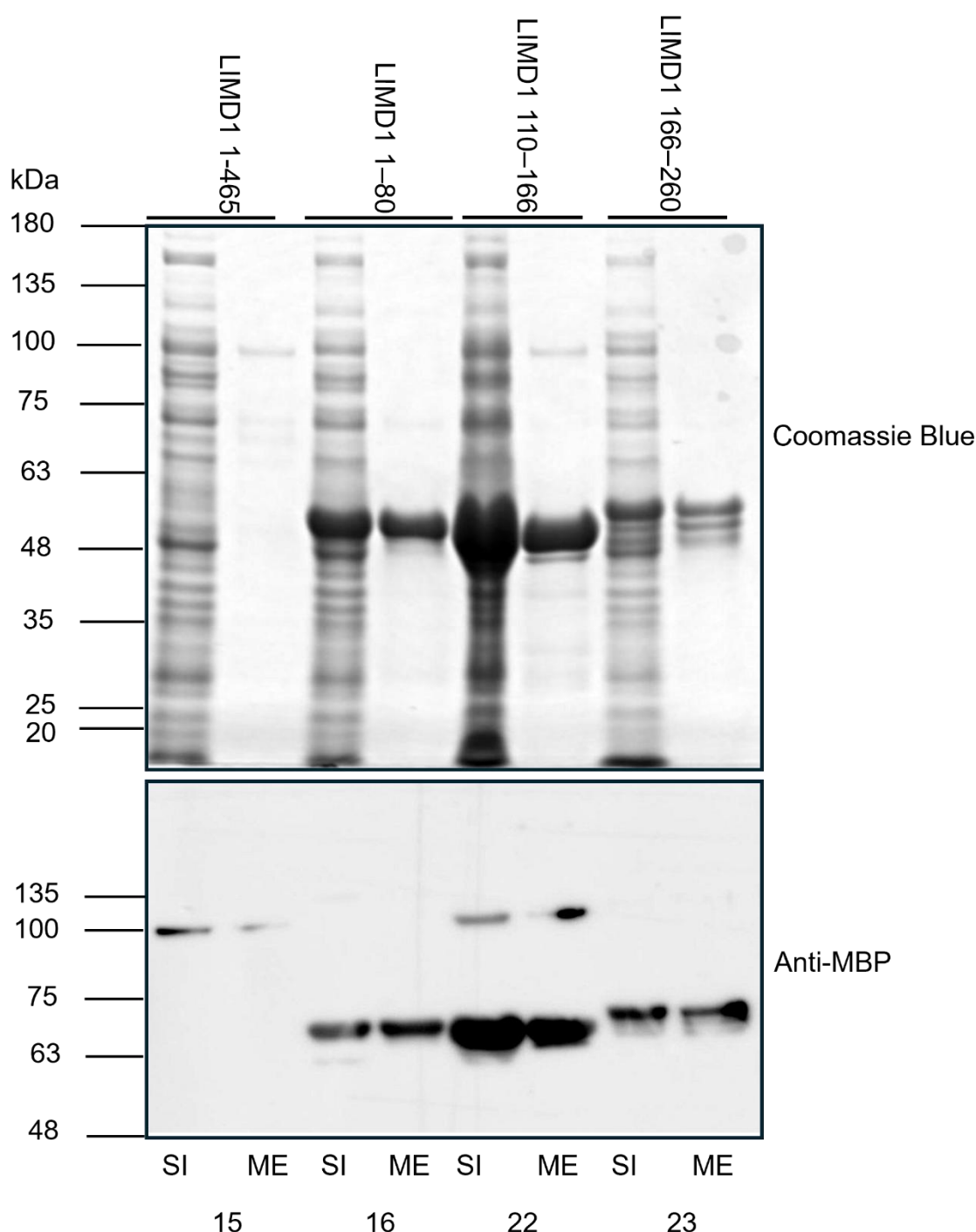


Figure 3.11. Magnetic nickel-affinity purification of LIMD1 constructs 15, 16, 22 and 23. Protein fractions were resolved by SDS-PAGE on 8% Bis-Tris gels. Upper panel: Coomassie Brilliant Blue-stained gel. Lower panel: anti-MBP western blot of the corresponding fractions. Constructs shown are LIMD1 1–465 (15), LIMD1 1–80 (16), LIMD1 110–166 (22), and LIMD1 166–260 (23). For each construct, sonicated input (SI) and magnetic bead elution (ME) fractions are shown following purification using magnetic nickel affinity beads under native conditions. Prestained protein molecular weight marker (10–245 kDa; Abcam) was used for molecular weight estimation.

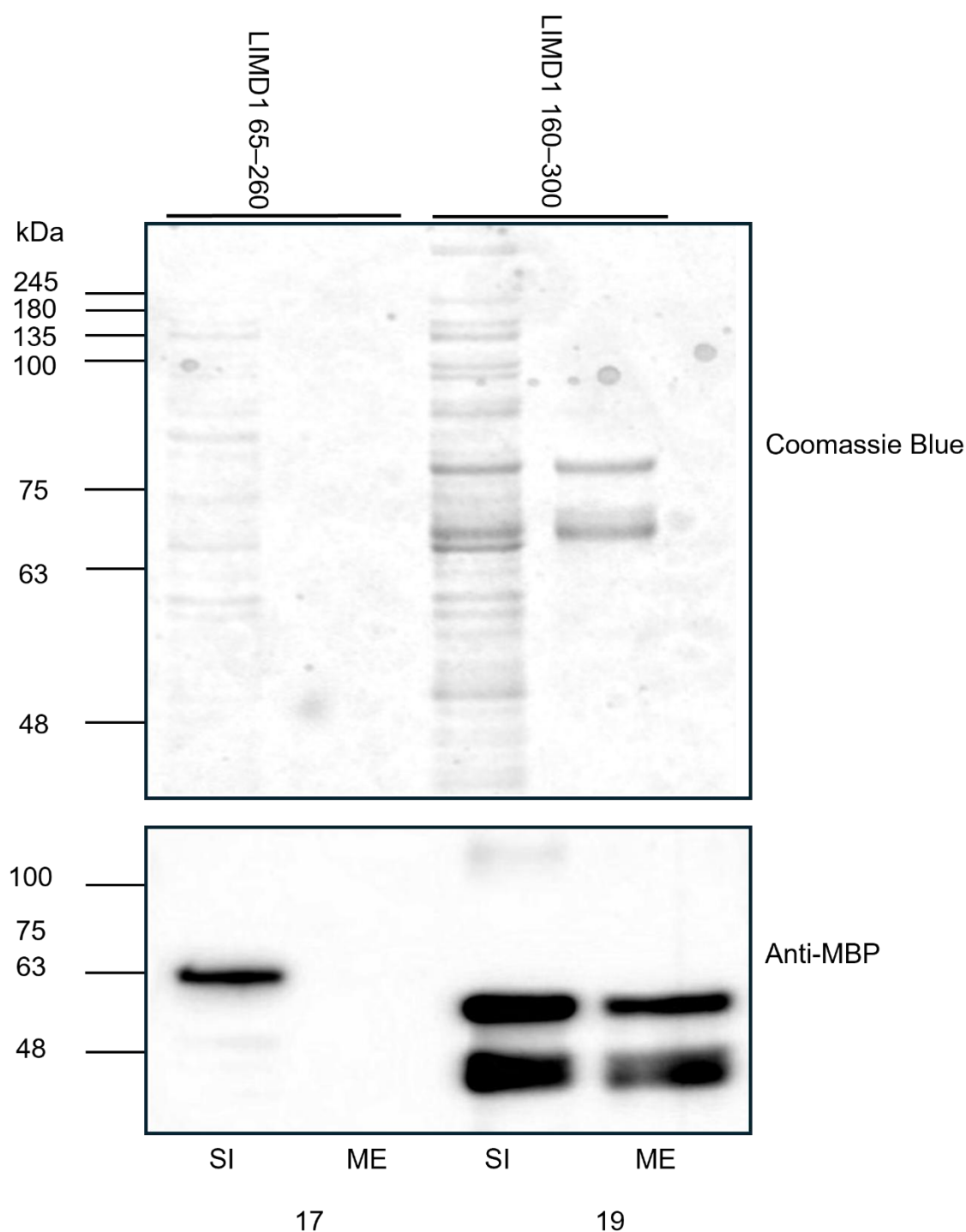


Figure 3.12. Magnetic nickel affinity bead purification of LIMD1 constructs 17 and 19. Protein fractions were resolved by SDS-PAGE on 8% Bis-Tris gels. Upper panel: Coomassie Brilliant Blue-stained gel. Lower panel: anti-MBP western blot of the corresponding fractions. Constructs shown are LIMD1 65–260 (17) and LIMD1 160–300 (19). For each construct, sonicated input (SI) and magnetic bead elution (ME) fractions are shown following purification using magnetic nickel affinity beads under native conditions. Prestained protein molecular weight marker (10–245 kDa; Abcam) was used for molecular weight estimation.

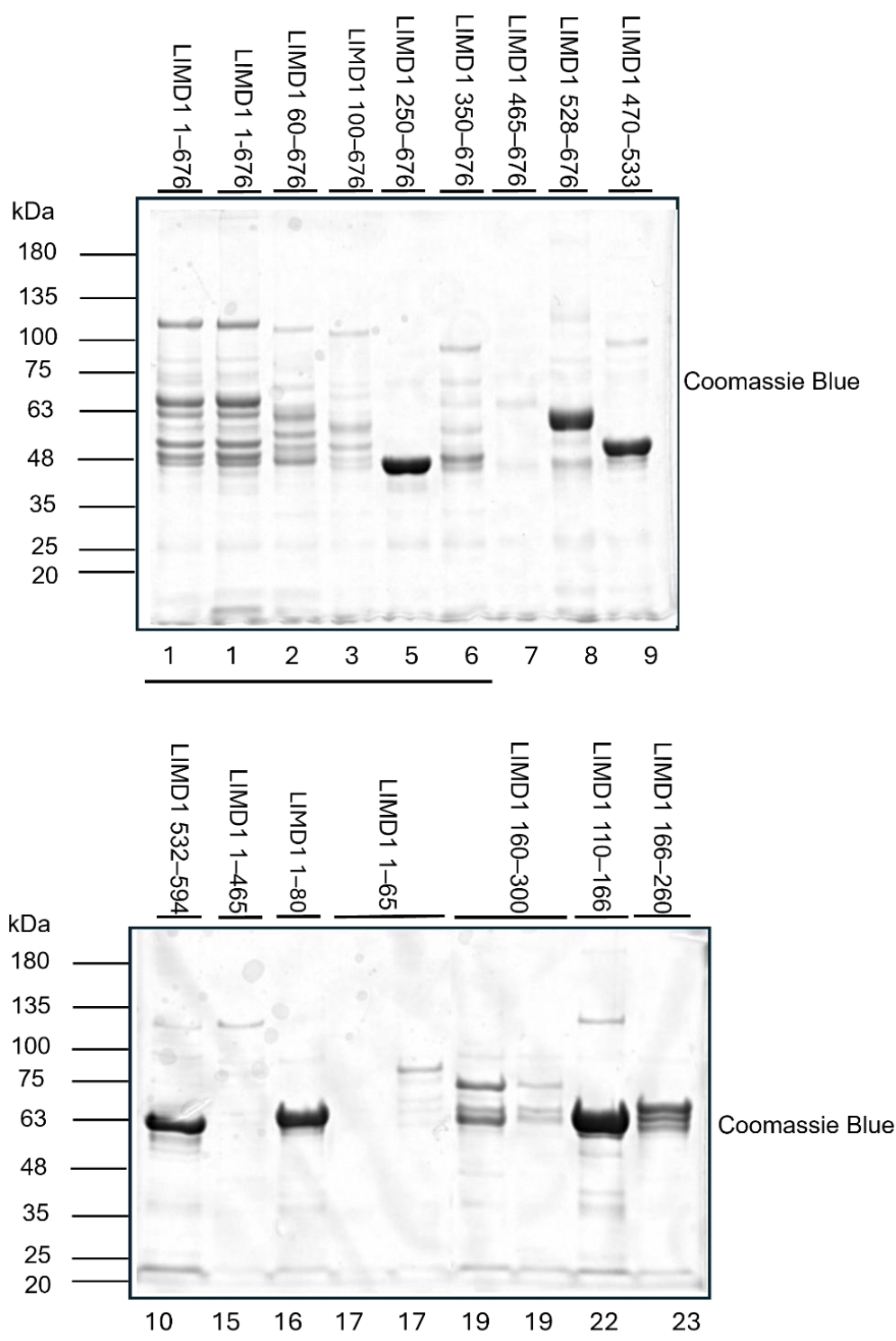


Figure 3.13. Coomassie-stained SDS-PAGE analysis of purified 6xHis-MBP-LIMD1 constructs. Elution fractions containing the indicated recombinant LIMD1 constructs were resolved by SDS-PAGE and visualised by Coomassie Brilliant Blue staining. Construct numbers are indicated below each lane, with the corresponding LIMD1 amino-acid boundaries shown above. The two full-length LIMD1 construct 1 preparation represents Ni^{2+} magnetic-bead purification following 1-h and overnight bead incubation, respectively. Horizontal lines beneath construct numbers denote samples purified using Ni^{2+} magnetic beads; samples without an underline were obtained by Ni^{2+} spin-column purification. Molecular masses of the protein marker are indicated in kDa.

3.8 LIMD1-Directed Amylose Affinity Pull-Down of Candidate Associated

Proteins

Full-length LIMD1 and constructs 2, 3, 7 and 15 were examined by amylose affinity pull-down using untransfected HEK293FT whole-cell lysate (Figure 3.14). Endogenous ARIH1 and AGO2 were readily detected in the input lysate, with weaker signals also present in the bead-only control and several LIMD1 pull-down fractions. Neither protein showed clear construct-selective enrichment relative to the bead-only control. PGAM5 was detected in the input and weakly in the bead-only control, with no clear enrichment in the LIMD1 pull-down fractions. TPX2 immunoblotting did not produce an interpretable specific signal under the conditions tested.

Anti-MBP and anti-6×His immunoblotting showed marked differences in bait recovery between constructs. Full-length LIMD1 produced the strongest immunoreactive signal, whereas constructs 2, 3, 7 and 15 showed weaker and variable recovery. These differences prevented direct comparison of candidate-protein enrichment between constructs. Under the conditions tested, no ARIH1-, AGO2- or PGAM5-associated LIMD1 region could therefore be assigned.

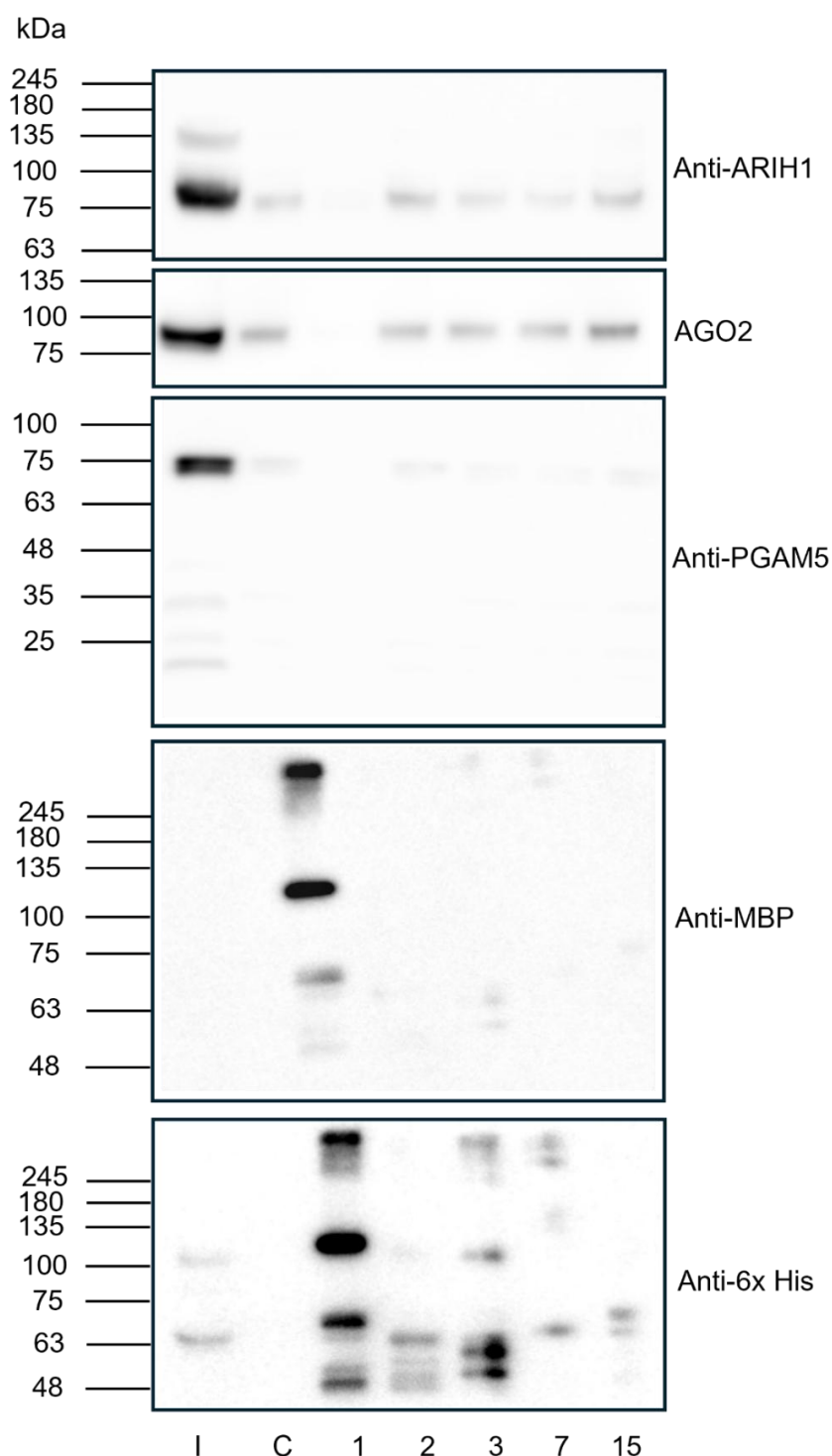


Figure 3.14. LIMD1-directed amylose affinity pull-down analysis of ARIH1, AGO2 and PGAM5. Recombinant 6×His–MBP–LIMD1 proteins were immobilised on amylose magnetic beads and incubated with untransfected HEK293FT whole-cell lysate. Western blotting was performed for ARIH1, AGO2 and PGAM5, while anti-MBP and anti-6×His immunoblotting assessed recovery of the recombinant LIMD1 bait proteins. Lanes represent lysate input (I), bead-only control (C), full-length LIMD1 construct 1 (1–676), construct 2 (60–676), construct 3 (100–676), construct 7 (465–676) and construct 15 (1–465). Molecular masses are indicated in kDa.

4. Discussion

4.1 Validity of the LIMD1 truncation approach

Deletion mapping requires each purified recombinant protein to represent its encoded LIMD1 sequence. Sequence verification confirms the plasmid but not protein integrity. This is important here because both 6×His and MBP are located at the N-terminus. A shortened protein could therefore retain both tags while lacking part of the LIMD1 sequence being tested. Detection with anti-6×His and anti-MBP antibodies cannot distinguish this from an intact fusion protein.

MBP is commonly used to improve soluble recombinant protein production. Kapust and Waugh showed that MBP was particularly effective at increasing soluble recovery of otherwise poorly soluble proteins.²⁰ However, soluble recovery does not establish correct folding of the attached protein. Raran-Kurussi and Waugh later used cleavage of His–MBP fusion proteins to determine whether the passenger protein remained soluble once MBP was removed.^{21,22}

Therefore, recovery of an MBP-tagged LIMD1 fragment does not establish that it behaves like the same region within full-length LIMD1. Protein integrity should therefore be confirmed before a construct is used to infer loss of binding. Detection of both ends of the recombinant protein would help distinguish intact fusion protein from N-terminal degradation products, while mass spectrometry would provide stronger confirmation where sufficient material is available. HRV 3C cleavage could also be used on selected constructs to determine whether their soluble recovery depends on MBP. These checks would reduce the risk that degradation or structural instability is interpreted as removal of an ARIH1-binding site.

4.2 Purification as a limitation to interaction mapping

Purification directly affects interaction mapping because it determines the amount and integrity of LIMD1 available as bait. A construct may be produced in bacteria but lost during lysis, clarification, affinity binding or elution. This makes recombinant expression and usable purified bait different experimental endpoints.

Nejadmoghaddam *et al.* compared nickel-coated magnetic matrices and found differences in binding capacity, protein loss and final recovery between matrices, with performance also depending on the recombinant protein being purified.²³

Their study used different proteins and matrices, so it cannot explain the behaviour of an individual LIMD1 construct. However, it supports the conclusion that the same 6×His purification method will not necessarily recover different fusion proteins equally. Using one purification method for every construct is therefore not necessarily the most controlled approach. A method may work well for one LIMD1 fragment and poorly for another. It would be more useful to select the purification condition that gives an intact preparation for each construct and then standardise the proteins before the binding experiment. The variable that needs to be matched is the amount of intact LIMD1 presented to ARIH1, not the purification route used to obtain it.

4.3 Testing whether the pull-down can map an ARIH1-binding region

Testing the hypothesis requires comparing ARIH1 association across comparable amounts of intact LIMD1 bait. Lower ARIH1 recovery could otherwise result from less available bait, while greater recovery could result from more bait rather than stronger binding. The present assay therefore could neither assign nor exclude an ARIH1-binding region. Future comparisons should use molar amounts of intact protein rather than equal sample volumes or equal protein mass, because the recombinant constructs differ substantially in molecular weight. The amount retained on the amylose beads should also be measured so that ARIH1 recovery can be related to the amount of bead-bound LIMD1.

The Appendix D densitometry describes individual blots but cannot estimate variation between independent experiments. Replicates would allow ARIH1 recovery to be normalised to bait and compared statistically. The control design should also separate different sources of background. A bead-only control identifies proteins retained by the amylose matrix, whereas an MBP-only control would identify proteins retained through the fusion partner. These controls answer different questions and should be included together in a repeated mapping experiment.

AGO2 offers a useful way to test whether the LIMD1 mapping system itself can detect a known regional interaction. Bridge *et al.* identified LIMD1 residues 140–166 as an AGO-binding region and showed that AGO2 recruitment is regulated by phosphorylation at Ser387. This provides a known regional interaction against which the truncation system can be validated.²⁴ However, total AGO2 in mammalian lysate may not provide an ideal positive control because its binding competence depends on phosphorylation state. A purified AGO2 preparation or defined AGO2 region with the appropriate binding state would therefore provide a stronger validation experiment.

The use of whole-cell lysate also limits what the ARIH1 pull-down can establish. Association from HEK293FT lysate may be direct or may be mediated by another cellular protein. In contrast, a purified LIMD1–ARIH1 assay can test whether the two proteins are sufficient for binding. These approaches should therefore be used sequentially rather than treated as interchangeable.

Reciprocal FLAG–ARIH1 capture would provide an additional test because recovery of the same LIMD1 region in the opposite orientation would reduce the chance that the result is specific to MBP or amylose beads. FLAG–ARIH1 protein expression should first be confirmed by anti-FLAG immunoblotting because sequence verification confirms the plasmid but not production of the protein.

4.4 Comparison with published LIMD1–ARIH1 studies

Published studies establish the existence and functional relevance of the LIMD1–ARIH1 pathway, whereas this study addressed where ARIH1 associates with LIMD1. The studies of Wu et al. and Xie et al. address a different question from the present work. Their experiments establish ARIH1-dependent PD-L1 ubiquitination using cellular and purified systems, whereas the present study aimed to determine which region of LIMD1 is required for ARIH1 association. The published catalytic evidence therefore cannot identify the LIMD1-binding site, but it defines the functional assay that should follow successful domain mapping.^{13,15} Shah et al. provide the most relevant benchmark because their purified-protein experiment establishes full-length LIMD1–ARIH1 binding, while leaving the interaction site unresolved.¹⁶ The present project asks whether that interaction depends on a defined part of LIMD1. Failure to map the region does not conflict with the published interaction because the main limitation was

comparison of unequal and incompletely validated recombinant baits rather than absence of ARIH1 from the system.

Shah *et al.* also found that ARIH1 depletion had a stronger effect on PD-L1 than LIMD1 loss. This suggests that ARIH1 can retain activity that does not depend entirely on LIMD1. Mapping the LIMD1-binding region would therefore identify the LIMD1-dependent part of ARIH1 regulation rather than define all ARIH1 control of PD-L1.¹⁶

Previous work on LIMD1 also suggests that more than one structural arrangement may be possible. Foxler *et al.* showed that LIMD1 can recruit components of the HIF degradation machinery through different regions, while Bridge *et al.* showed that AGO2 and TNRC6A use distinct LIMD1 interfaces.^{11,24} ARIH1 may therefore bind one discrete sequence, more than one LIMD1 region, or a surface that depends on the arrangement of full-length LIMD1. This possibility should be considered if isolated N- and C-terminal regions both fail to reproduce full-length binding.

4.5 Strategy for identifying the ARIH1-binding region

Mapping should first distinguish the pre-LIM and C-terminal LIM-containing regions. If association is retained within the C-terminal region, individual and combined LIM-domain constructs can narrow the site. If it remains within the pre-LIM region, smaller internal fragments can be tested. If neither region reproduces full-length binding, an overlapping construct spanning the boundary should be examined before concluding that full-length structure is required.

Binding of an isolated fragment would show that the region is sufficient under the assay conditions, but would not establish that it is required in full-length LIMD1. Bridge *et al.* addressed this problem in the AGO2 system using an internal LIMD1 deletion after regional mapping.²⁴ A similar internal deletion or selected mutation should therefore be introduced into full-length LIMD1 once an ARIH1-associated region has been

identified. Loss of ARIH1 association would provide stronger evidence that the sequence is required while leaving the remainder of LIMD1 intact. A purified two-protein experiment should then test direct binding between ARIH1 and the candidate LIMD1 region. If association is retained with purified proteins, the two proteins are sufficient for the interaction. If binding occurs only in lysate, another cellular factor may contribute.

More quantitative methods such as biolayer interferometry or surface plasmon resonance would only become useful after a reproducible direct interaction has been established. Their role would be to compare binding affinity between wild-type and altered LIMD1 proteins, rather than to solve the initial problem of protein integrity or unequal bait recovery.

4.6 Biological significance in NSCLC

The main value of mapping the interaction is to determine how much of the PD-L1 phenotype caused by LIMD1 loss depends specifically on ARIH1 binding. A binding-deficient full-length LIMD1 mutant could test whether loss of ARIH1 association is sufficient to impair PD-L1 ubiquitination and stability in lung cancer cells. LIMD1 supports ARIH1-dependent PD-L1 ubiquitination and degradation; therefore, disrupting the LIMD1–ARIH1 interaction could favour PD-L1 accumulation rather than produce a beneficial effect. Wild-type and binding-deficient LIMD1 could be introduced into LIMD1-deficient lung cancer cells and compared for ARIH1–PD-L1 association, PD-L1 ubiquitination and PD-L1 stability. If only wild-type LIMD1 restored these effects, the mapped interaction would be required for this aspect of LIMD1 tumour-suppressive function. If both proteins behaved similarly, ARIH1 binding alone would not explain the PD-L1 phenotype.

Previous work also suggests that LIMD1 loss can create tumour-specific vulnerabilities. Davidson *et al.* reported a significant relationship between LIMD1 expression and sensitivity to PF-477736 in LUAD cells, with a Pearson correlation of $r = 0.579$, $P = 0.0302$.²⁵ This does not imply that ARIH1 should be targeted in the same way. Instead, it shows that loss of a scaffold protein can produce measurable biological states that may be used to distinguish LIMD1-deficient tumours.

5. Conclusion

This study built a sequence-verified LIMD1 truncation platform and established workable, though construct-dependent, expression and purification conditions. The pull-down did not resolve an ARIH1-, AGO2- or PGAM5-associated region because unequal, non-normalised bait recovery prevented reliable comparison between constructs. The findings are therefore inconclusive and do not demonstrate absence of binding. The appropriate next step is to reproduce the published full-length interaction using purified, quantified proteins before repeating domain mapping and testing candidate regions in ARIH1-dependent PD-L1 ubiquitination assays.

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7. Appendices

Appendix A — Supplementary Experimental Methods

Table A1. Q5 mutagenesis PCR reaction composition.

Component	Stock concentration	Volume per reaction
Q5 High-Fidelity 2× Master Mix	2×	12.5 μ L
Forward primer	10 μ M	1.25 μ L
Reverse primer	10 μ M	1.25 μ L
Template plasmid DNA	—	1.0 μ L
Nuclease-free water	—	9.0 μ L
Total reaction volume		25.0 μ L

Construct 15 template DNA was diluted 1:50 before amplification. Primer sequences are provided in Appendix Table B1.

Table A2. Thermal-cycling programmes used for site-directed mutagenesis.

Stage	LIMD1 Construct 15-1	FLAG–ARIH1
Initial denaturation	98°C, 30 s	98°C, 30 s
Denaturation	98°C, 10 s	98°C, 10 s
Annealing	67°C, 15 s	59°C, 15 s
Extension	72°C, 3 min 30 s	72°C, 3 min 40 s
Number of cycles	30	30
Final extension	72°C, 2 min	72°C, 2 min
Hold	4°C	4°C

The LIMD1 reaction removed the terminal stop codon from Construct 15 to regenerate full-length LIMD1. The FLAG–ARIH1 reaction replaced the N-terminal 6×His coding sequence with a FLAG epitope.

Table A3. KLD reaction composition.

Component	Volume
Mutagenesis PCR product	1.0 μ L
KLD Reaction Buffer	2.5 μ L
KLD Enzyme Mix	0.5 μ L
Nuclease-free water	1.5 μ L
*Recorded total volume	5.5 μ L

*The reaction was mixed gently and incubated for 5 min. A 2.5 μ L aliquot was subsequently transformed into 25 μ L high-efficiency DH5 α cells.

Table A4. Buffer compositions used for protein purification, cell lysis and immunoblotting

Buffer	Composition
NPI-10	50 mM NaH ₂ PO ₄ , 300 mM NaCl and 10 mM imidazole, pH 8.0
NPI-20	50 mM NaH ₂ PO ₄ , 300 mM NaCl and 20 mM imidazole, pH 8.0
NPI-500	50 mM NaH ₂ PO ₄ , 300 mM NaCl and 500 mM imidazole, pH 8.0
Detergent-assisted bacterial lysis buffer	NPI-10 containing 1% NP-40
HEK293FT NP-40 cell-lysis buffer	50 mM Tris-HCl, pH 8.0, 150 mM NaCl and 1% NP-40, supplemented immediately before use with protease-inhibitor cocktail
Amylose pull-down buffer	PBS containing 0.1% Triton X-100 and 5% glycerol
PBST	PBS containing 0.05% Tween-20
Blocking buffer	5% (w/v) non-fat milk prepared in PBST
Wet-transfer buffer	1 $\times$ Tris-glycine containing 10% (v/v) methanol

Table A5. Native Ni-NTA spin-column purification procedure

Stage	Procedure
Pellet preparation	Bacterial cells from 50 mL expression cultures were harvested at 3,000 rpm for 15 min. Pellets were stored at -80°C until purification.
Resuspension	Each pellet was resuspended in 1.4 mL NPI-10 supplemented with 14 μ L lysozyme (50 mg/mL) and 1 μ L Benzonase (25 U/ μ L).
Lysis	Samples were incubated on ice for 15–30 min.
Clarification	Lysates were centrifuged at $12,000 \times g$ for 15–30 min at 4°C . An aliquot of clarified lysate was retained as the input fraction.
Column equilibration	Each Ni-NTA spin column was equilibrated with 600 μ L NPI-10 and centrifuged at $890 \times g$ for 2 min.
Sample loading	Clarified lysate was applied in two successive 600 μ L aliquots. Each aliquot was centrifuged at $270 \times g$ for 5 min, and the flow-through was retained.
Washing	Columns were washed twice with 600 μ L NPI-20 and centrifuged at $890 \times g$ for 2 min after each wash.
Elution	Recombinant protein was eluted with 300 μ L NPI-500 by centrifugation at $890 \times g$ for 2 min.
Fraction analysis	Input, flow-through, wash and elution fractions were retained where required for SDS-PAGE, Coomassie staining and immunoblotting.

Table A6. Magnetic Ni-NTA purification procedure

Stage	Procedure
Lysate preparation	Bacterial pellets were resuspended in 1.4 mL NPI-10 containing 14 μ L lysozyme (50 mg/mL) and 1 μ L Benzonase (25 U/ μ L).
Sonication	Samples were sonicated on ice at 25% amplitude using 10 s on/30 s off cycles to a cumulative pulse-on time of 90 s.
Clarification	Lysates were centrifuged at 12,000 $\times$ g for 15–30 min at 4°C.
Bead quantity	50 μ L HisPur Ni-NTA Magnetic Beads per construct
Equilibration	Beads were equilibrated twice with 500 μ L NPI-10.
Sample loading	A 300 μ L aliquot of clarified sonicated lysate was added to each bead preparation.
Binding	The initial full-length purification was incubated for 1 h. Subsequent purifications were incubated overnight at 4°C with end-over-end rotation.
Flow-through collection	Beads were collected magnetically, and unbound lysate was retained as the flow-through fraction.
Washing	Beads were washed four times with 1 mL NPI-20, with complete resuspension during each wash.
Elution	Bound protein was eluted with 200 μ L NPI-500 during 5 min of rotation.
Fraction analysis	Input, flow-through and elution fractions were analysed by SDS–PAGE, Coomassie staining and anti-MBP immunoblotting.

Table A7. FLAG–ARIH1 transfection and HEK293FT lysate preparation

Stage	Procedure
Culture conditions	HEK293FT cells were maintained in DMEM containing 10% FBS, 1% GlutaMAX and 1% penicillin–streptomycin at 37°C and 5% CO ₂ .
Final culture format	Two 10 cm HEK293FT dishes
Plasmid DNA	5 μ g FLAG–ARIH1 plasmid DNA per dish
Transfection reagent	15 μ L DharmaFECT KB per dish
Transfection medium	Opti-MEM; final transfection-mixture volume of 400 μ L per dish
Complex formation	DNA and DharmaFECT KB were diluted separately and incubated for 5 min. The solutions were combined and incubated for a further 20 min.
Transfection	Complexes were added dropwise to the cells, which were incubated for 24 h at 37°C and 5% CO ₂ .
Washing	Cells were washed once with PBS.
Lysis	Each dish was lysed in 200 μ L NP-40 cell-lysis buffer for 5 min on ice.
Lysis-buffer composition	50 mM Tris–HCl, pH 8.0, 150 mM NaCl and 1% NP-40, supplemented with protease inhibitors
Collection	Cells were collected using a sterile cell scraper.
Clarification	Lysates were centrifuged at 12,000 $\times$ g for 5 min at 4°C.
Final pull-down lysate	Lysates were prepared for reciprocal FLAG–ARIH1 affinity capture; protein expression and reciprocal pull-down analysis were not completed.

Table A8. Amylose magnetic-bead pull-down procedure

Stage	Procedure
Experimental conditions	Bead-only control and recombinant LIMD1 Constructs 1, 2, 3, 7 and 15
Pull-down buffer	PBS containing 0.1% Triton X-100 and 5% glycerol
Bead quantity	15 μ L settled Amylose Magnetic Beads per reaction
Initial bead wash	Beads were washed once with 300 μ L pull-down buffer.
Bead resuspension	Washed beads were resuspended in 100 μ L pull-down buffer.
Bait addition	Twenty microlitres of the appropriate LIMD1 elution was added to each construct reaction. No recombinant protein was added to the bead-only control.
Bait immobilisation	Reactions were incubated for 1 h at 4°C with end-over-end rotation.
Input preparation	An 80 μ L aliquot of HEK293FT lysate was mixed with 20 μ L 5 $\times$ Laemmli sample buffer and heated at 95°C for 5 min.
Lysate addition	A 160 μ L aliquot of untransfected HEK293FT whole-cell lysate was added to each pull-down reaction.
Interaction incubation	Reactions were incubated overnight at 4°C with end-over-end rotation.
Washing	Beads were washed three times with 300 μ L pull-down buffer.
Elution	Bound material was recovered in Laemmli sample buffer by heating at 95°C for 5 min.
Immunoblot analysis	Recovery of ARIH1, AGO2, PGAM5 and TPX2 was assessed alongside MBP and 6 $\times$ His bait detection.
Experimental replication	The pull-down was performed once.
Control limitations	A bead-only control was included, but an MBP-only control was not included.
Bait loading	Equal LIMD1 elution volumes were used; bait quantities were not molar-normalised.

Table A9. SDS–PAGE, Coomassie staining and immunodetection procedure

Stage	Procedure
Sample preparation	Twenty microlitres of protein sample was mixed with 5 μ L 5 $\times$ Laemmli sample buffer.
Denaturation	Samples were heated at 95°C for 5 min and briefly centrifuged.
Sample loading	Ten microlitres of prepared sample was loaded per lane unless otherwise stated.
Gel composition	8% resolving gel with a 5% stacking gel
Electrophoresis	80 V for 10 min, followed by 125 V for approximately 50 min
Initial gel wash	Gels selected for total-protein staining were washed in distilled water for 5 min.
Fixation	Gels were incubated in fixation buffer for 10 min with gentle agitation.
Coomassie staining	Gels were stained for 10 min, microwaved for 20 s and incubated in staining solution for a further 10 min.
Destaining	Gels were destained in fresh fixation buffer for approximately 30 min, replacing the solution as required until the background was clear.
Imaging of stained gels	Coomassie-stained gels were imaged using a Bio-Rad ChemiDoc system.
Membrane preparation	PVDF membranes were activated in methanol before transfer.
Protein transfer	Proteins were transferred in 1 $\times$ Tris–glycine buffer containing 10% methanol at 90 V for 70 min.
Blocking	Membranes were blocked in 5% non-fat milk in PBST for 1 h at room temperature.
Primary-antibody incubation	Membranes were incubated with the appropriate primary antibody overnight at 4°C. Working dilutions are provided in main-text Table 2.3.
Membrane washing	Membranes were washed in PBST after primary- and secondary-antibody incubations.
Secondary-antibody incubation	The appropriate HRP-conjugated secondary antibody was applied for 1 h at room temperature.
Chemiluminescent detection	Signal was developed using SuperSignal West Pico PLUS substrate and imaged using a Bio-Rad ChemiDoc system.

Appendix B — Oligonucleotide Primers

Primer	Sequence (5'–3')	Length	Order no.	reference no.
464K.AAG.1_FWD	TGCTCACCCGAAGGCTGATTA CT	23 nt	21529244	515938642
464K.AAG.1_REV	TCCATCTCTCGCTCCAGAC	19 nt	21529244	515938643
6HisToFLAG ARIH1 F	GATGATGATAAAGCTAGCGGA ATGGAC	27 nt	21539174	516084827
6HisToFLAG ARIH1 R	ATCTTTATAATCCATGAATTCA GCACAC	28 nt	21539174	516084828

Table B1. Oligonucleotide primers used for site-directed mutagenesis in this study. Oligonucleotides were synthesised by Integrated DNA Technologies (IDT). The 464K.AAG.1 primer pair was used during regeneration of the full-length LIMD1 construct, while the 6HisToFLAG ARIH1 primer pair was used to replace the N-terminal 6×His tag of ARIH1 with a FLAG epitope.

Appendix C — Sanger Sequencing Results

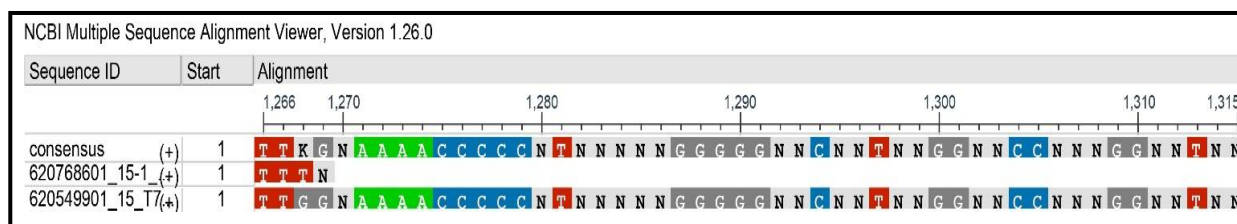


Figure C1. Sanger sequencing verification of regenerated full-length LIMD1 construct 1. Construct 1 was regenerated from construct 15 by site-directed mutagenesis. Sequence alignment confirmed the expected full-length LIMD1 sequence following regeneration.

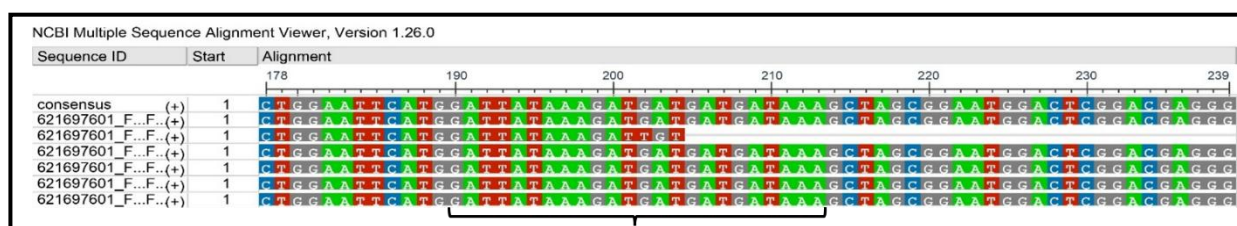


Figure C2. Sanger sequencing verification of FLAG-tag insertion in ARIH1. Site-directed mutagenesis was performed to replace the N-terminal 6×His tag with a FLAG tag. Alignment of six Sanger sequencing reads (FLAG-ARIH1_F1–F6) confirmed the expected FLAG-tag sequence. The F2 read showed reduced sequence quality beyond the modified region, resulting in a shorter alignment.

Appendix D — Supplementary Recombinant LIMD1 Expression Data

Table D1. Densitometric analysis of anti-6×His immunoblotting of LIMD1 constructs 2–10. Densitometry was performed using ImageJ. Raw integrated density (RawIntDen) values were normalised to construct 8, which showed the highest signal on this membrane and was assigned 100%. Values are descriptive measurements from a single immunoblot.

Construct	Expected MW (kDa)	RawIntDen	Relative expression (%)
2	111.2	35,917,431	47.2
3	106.8	31,038,491	40.8
5	90.3	20,737,414	27.3
6	79.3	51,624,442	67.9
7	66.6	51,732,747	68.0
8	59.7	76,074,284	100.0
9	50.3	66,787,937	87.8
10	50.2	67,975,732	89.4

Table D2. Densitometric analysis of anti-6×His immunoblotting of LIMD1 constructs 11–23. Densitometry was performed using ImageJ. RawIntDen values were normalised to construct 11, which showed the highest signal on this membrane and was assigned 100%. Values are descriptive measurements from a single immunoblot.

Construct	Expected MW (kDa)	RawIntDen	Relative expression (%)
11	52.0	97,006,625	100.0
12	57.1	66,229,027	68.3
13	58.5	78,217,265	80.6
14	65.3	59,827,226	61.7
15	94.5	26,172,763	27.0
16	52.1	74,740,233	77.0
17	64.9	64,676,029	66.7
19	58.8	46,174,622	47.6
22	49.6	12,678,360	13.1
23	53.8	35,734,884	36.8

Table D3. Densitometric analysis of anti-MBP immunoblotting of LIMD1 constructs 2–10. Densitometry was performed using ImageJ. RawIntDen values were normalised to construct 10, which showed the highest signal on this membrane and was assigned 100%. Values are descriptive measurements from a single immunoblot.

Construct	Expected MW (kDa)	RawIntDen	Relative expression (%)
2	110.4	51,407,429	97.0
3	106.0	41,289,413	77.9
5	89.5	14,678,620	27.7
6	78.5	46,314,370	87.4
7	65.8	50,500,554	95.3
8	58.9	42,095,822	79.4
9	49.5	43,367,868	81.8
10	49.4	52,984,629	100.0

Table D4. Densitometric analysis of anti-MBP immunoblotting of LIMD1 constructs 11–23. Densitometry was performed using ImageJ. RawIntDen values were normalised to construct 23, which showed the highest signal on this membrane and was assigned 100%. Values are descriptive measurements from a single immunoblot.

Construct	Expected MW (kDa)	RawIntDen	Relative expression (%)
11	51.2	59,087,654	90.0
12	56.3	38,234,076	58.2
13	57.7	36,632,815	55.8
14	64.5	48,147,338	73.3
15	93.7	37,182,495	56.6
16	51.3	23,810,404	36.3
17	64.1	30,481,670	46.4
19	58.0	47,029,118	71.6
22	48.8	51,991,605	79.2
23	53.0	65,655,878	100.0

Table D5. Densitometric analysis of anti-MBP immunoblotting of the selected LIMD1 construct panel. Densitometry was performed using ImageJ. RawIntDen values were normalised to construct 8, which showed the highest signal on this membrane and was assigned 100%. Values are descriptive measurements from a single immunoblot.

Construct	Expected +MBP MW (kDa)	RawIntDen	Relative expression (%)
15-1 / Construct 1	116.9	34,137,537	49.5
2	110.4	36,903,458	53.5
3	106.0	18,793,471	27.2
4	99.4	41,742,788	60.5
5A	89.5	47,204,269	68.4
5B	89.5	36,432,444	52.8
5C	89.5	40,289,607	58.4
5D	89.5	39,693,236	57.6
6	78.5	19,969,673	29.0
7	65.8	27,757,605	40.2
8	58.9	68,969,026	100.0

Table D6. Densitometric analysis of anti-LIMD1 immunoblotting of full-length and N-terminally truncated LIMD1 constructs. Densitometry was performed using ImageJ. RawIntDen values were normalised to regenerated full-length construct 1 (15-1), which showed the highest signal and was assigned 100%. Values are descriptive measurements from a single immunoblot.

Construct	LIMD1 MW (kDa)	RawIntDen	Relative signal (%)
15-1 / Construct 1	74.4	79,516,802	100.0
2	67.9	73,389,248	92.3
3	63.5	63,740,920	80.2
4	56.9	37,512,726	47.2